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Monetary Policy and Stock Prices: Evidence from the UK and Norway

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Abstract

This dissertation examines the impact of monetary policy on the stock market in the UK and Norway, utilising Structural Vector Autoregression (SVAR) modelling on a monthly dataset spanning from January 1997 to December 2023. We study the effects of domestic and foreign monetary policy while controlling for macroeconomic factors. We find that a contractionary monetary policy shock has a negative impact on real equity returns and remains consistent across models. We find that foreign policy spillovers are heterogeneous, as Norwegian equities exhibit an early decline, whereas the UK responses are insignificant in the short run and have delayed effects that are more persistent. We also find that monetary policy explains a small share of equity volatility at business cycle horizons. The findings align with the consensus that monetary policy affects equities on impact primarily through the discount rate channel, but it is a blunt tool for managing stock markets.

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Chapter 1: Introduction

1.1 Research objectives

Equity prices are a primary transmission channel from monetary policy to the real economy; they shape firms' costs of funding and investment, household wealth and spending, and the pricing of risk across markets. For informed policymaking, understanding the timing, persistence, and direction of equity responses is crucial for communication, evaluation, and financial stability monitoring. In modern asset pricing, equity values are calculated as the discounted stream of expected real cash flows. The stochastic factor incorporates both the anticipated path of real short rates and risk premia; therefore, monetary policy impacts valuations directly through the discount rate channel (Cochrane, 2011).

Following the post-COVID-19 period, inflation surge and sharp global tightening, uncertainty regarding the path of monetary policy and risk premia has increased (Rathnayaka, 2024), serving as a contemporary measure of how much equity variation is attributable to domestic monetary policy versus global factors. Event-study and announcement window designs show that unexpected tightening generally reduces equity valuation, and heteroskedasticity-based identification highlights a significant immediate effect (Bernanke and Kuttner, 2005). However, measured effects are sensitive to identification, especially when announcements disclose information about future outlooks.

Modern empirical work in the UK context often employs high-frequency instruments, but equity responses are frequently studied within domestic systems (Cesa-Bianchi, Thwaites and Viconda, 2020). For Norway, comparable evidence is more limited, and combining these two advanced open economies is rarely analysed together using a common monthly SVAR with short-run restrictions. Moreover, research on small open economies shows that foreign monetary policy influences financial conditions domestically; therefore, its omission can bias the measured domestic effect (Hausman and Wongswan, 2011). Recent studies have examined this effect in small open economies heavily dependent on the US, such as Canada and Mexico, highlighting the importance of foreign policy spillovers (Li, Iscan & Xu, 2010; Alba & Carrillo, 2024).

To address these gaps, the dissertation takes a comparative approach using consistent economic controls, restrictions and data to assess the impact of domestic monetary policy on real equity returns in open economies, to show how standard modelling choices, especially in the inclusion of foreign policy rates and recursive compared to short-run restricted models, shape those estimates.

The research aims to:

1. Estimate the sign, timing and persistence of domestic shocks on national equity indices in the UK and Norway, using monthly SVAR modelling.
2. Identify the effect of foreign policy on domestic equity valuations.
3. Compare identification choices in recursive and short-run ordering and restrictions.
4. Address concerns of restriction and lag-dependent estimations by providing robustness models.
5. Policy implications based on the results of our findings.

This study contributes to the monetary-policy and asset-pricing literature by considering open economy modelling to identify the distinct effects of domestic and foreign policy shocks on equities, and by explicitly testing recursive and non-recursive short-run restrictions. The UK and Norway comparative design highlights how policy transmission varies across a market-based system and a more bank-centred small open economy, while addressing key identification concerns, such as omitted foreign shocks, using accumulated IRFs, bootstrap confidence intervals.

1.2 Structure of the dissertation

This dissertation is organised in seven chapters examining how domestic and foreign monetary policy shape UK and Norwegian equities. Chapter 1 states the questions, motivation and contributions; Chapter 2 reviews theory and evidence on transmission. Chapter 3 sets out the identification strategy and robustness; Chapter 4 covers data, transformations, and diagnostics. Chapter 5 presents IRFs/FEVDs for domestic and foreign shocks; Chapter 6 interprets the results, explores heterogeneity and robustness, and notes limitations and policy implications. Chapter 7 concludes with our findings.

Chapter 2: Literature Review

2.1 Overview of monetary policy

The studies on monetary policy's channels of impact have undergone rigorous analysis since the "expectations revolution" (Miller, 1994), which overturned the idea of a permanent inflation adjustment and the trade-off with lower unemployment. Friedman (1968) and Phelps (1967) suggested that once households and firms form rational expectations, surprise inflation shocks do not systematically reduce unemployment; instead, monetary policy influences agents' future expectations of real interest rates and inflation. It is the changes in expectations that influence concurrent investment and consumption; therefore, only changes that impact future expected policies generate sustained effects on output and unemployment (Phelps 1967).

Following the theoretical deviations from the stable Phillips curve, Lucas (1976) demonstrates that as behaviour shifts when a monetary policy rule changes, reduced form correlations are no longer policy-invariant as forward-looking agents adjust their expectations and behaviour. The regime differences and their effects on policy implementation are highlighted by the discretionary tools policymakers employ to construct periodic rates, which invite time inconsistency, an inflation bias without lasting real gains. Barro & Gordon (1983) formalise this argument by indicating that when a policymaker values having the output above the natural level, they will be tempted to create surprise inflation, and anticipatory agents will raise their inflation expectations. Therefore, surprises disappear in the long run with no gain in real output. This inflation bias is therefore an equilibrium outcome of rational expectations.

The New-Keynesian Phillips curve is forward-looking; current inflation depends on expected future inflation and real marginal cost. Structural estimates in Galí & Gertler (1999) show that marginal costs are quantitatively important and that forward-looking behaviour explains much of inflation dynamics, evidence that monetary policy works chiefly by shaping expectations. Coupling an IS curve that links the output gap to the expected real rate with a forward-looking Phillips curve rationalises greater macro stability under predictable, Taylor-type rules. In open economies, the Mundell–Fleming framework highlights the policy assignment and exchange-rate-regime trade-offs under high capital mobility.

Monetary transmission moves through revisions to the expected path of medium-horizon real rates; when credibility anchors those expectations, real effects on spending and pricing persist.

Discretion instead leads to a predicted inflation bias, weakening traction on real activity. In open economies, the policy target governs how much imported inflation and exchange-rate pass-through are internalised; therefore, external conditions can transmit even when the local short rate barely moves.

2.2 Key theories: The role of monetary policy transmission on equity prices

Asset pricing identity

Under the present-value identity, monetary policy influences equity prices either by altering required returns (the discount-rate channel) or by modifying expected dividends (the cash-flow channel) Therefore, any unexpected movements in the share price should reflect news about the dividend stream or the discount rates.

Empirical event studies confirm the dominance of discount rate news; for example, Bernanke & Kuttner (2005), using Federal Funds data around policy announcement windows (1989-2002), show that an unanticipated 25 basis point easing increases the CRSP value-weighted index by about 1% on the same day. After filtering out dividends, nearly the entire effect is attributed to lower required returns rather than higher expected cash flows. Consequently, in analysing the error variance of the impact of monetary policy shocks, distinguishing policy from informational effects is important. Most evidence indicates that required returns, not dividend expectations, have been the primary factor that changed.

A considerable variance in the equity prices originating from monetary policy signals discount factor repricing and not sudden revisions to cash flows. Under the efficient market hypothesis, market prices reflect publicly available information. Malkiel (1989) emphasises that only unanticipated news can move asset prices in such markets. Having separated price moves into discount rate and cash flow news, we move on to contemporaneous impact repricing via the discount rate, and the lagged operating through demand, earnings, and inflation

Decomposition: Direct discount-rate and indirect cash-flow

Extending the Campbell & Ammar (1993) present value decomposition, we see the direct and indirect effects of discounted cash flow by making the transmission channel explicit. Monetary policy can move prices directly by changing the discount rate investors apply to cash flows and indirectly alter future cash flows via real economic transmissions. The direct effect links a policy-induced shift in market interest rates and returns to an immediate repricing of equities, and the indirect effect operates through activity and profits; therefore, it is slower and typically shows its effects beyond the announcement window.

Monetary policy also operates through spending, production, and profits; a tighter policy slows activity and earnings, so cash-flow effects emerge with lags that extend beyond the announcement window. Across 13 OECD economies (1972–2002), contractionary policy is associated with lower stock returns and looser policy with higher returns, consistent with an on-impact discount-rate effect and slower cash-flow adjustments through real activity (Ioannidis and Kontonikas, 2008).

Interest rates channel linking policy rates to aggregate demand and inflation

Monetary policy initially influences real short-term interest rates. In the New-Keynesian IS curve, current demand equals expected future demand minus inter-temporal substitution (Clarida, Gali & Gertler 1999). A surprise contractionary monetary policy that raises the policy rate above its natural level prompts households to postpone consumption in order to reduce the output gap. Macro evidence supports this theory; a VAR model covering data post-1980s found that a 100 basis point increase in the funds rate had a negative impact on real activity for 4-6 quarters and decreased core inflation by 0.4 percentage points over a 10-12 quarter horizon (Boivin, Kelly & Mishkin 2010).

As the Phillips curve translates the weaker output gap into lower inflation (Woodford 2003), the real-rate channel is the foundation that connects other equity-focused channels in our review to prices and activity. Monetary policy changes, guidance, and Quantitative Easing (QE) and their spillovers into foreign rates and credit spreads change the required return investors use to value equities, as the discount rate is composed of the expected risk-free path and equity risk premium, so equity prices react at once. It only propagates to the real economy if the change in effective real cost of funds makes households retime spending, therefore demand profits and their inflation moves. If the spending does not shift, the shock stays in the asset valuation, and the cash-flow effect is muted.

Intermediary risk premium

Monetary policy moves discount factors by shifting the price of risk borne by financial intermediaries. Easier policy relaxes capital constraints, compresses spreads, and lowers the equity risk premium; tighter policy does the reverse; therefore, equities reprice on announcement through the discount-rate component (Borio & Zhu, 2012). In intermediary-asset-pricing terms, a fall in aggregate intermediary capital raises the market equity premium, tightening the discount factor channel (He & Krishnamurthy, 2013).

A second layer operates through household asset demand. Lower safe rates induce portfolio substitution out of money or short bonds into equities, classic Mundell–Tobin logic (Mundell, 1963; Tobin, 1969). Modern evidence suggests that the rebalancing following rate changes is concentrated among stock-market participants, particularly higher-wealth households. Therefore, the strength of this channel depends on participation and wealth concentration (Melcangi & Sterk, 2020).

These mechanisms can reinforce or offset each other depending on who holds risk and how quickly they reallocate. In bank-centred, small open markets such as Norway, with exposure to cyclical and commodity shocks, the risk-taking channel predicts larger, more state-dependent equity responses to policy surprises than in more institutionally diversified markets like the UK. Empirically, this is borne out by studies interacting high-frequency policy shocks with bank capital and by comparing firm responses by risk-bearing capacity and financing constraints (Altavilla, Giannone & Lenza, 2016; Adrian, Etula & Muir, 2014).

Target and path channel of the information effect

Using daily identification, Gürkaynak, Sack and Swanson (2005) show that FOMC announcements load on two orthogonal shocks: a target factor (the surprise in the current policy rate) and a path factor (the revision to the expected path of short rates over the coming quarters). The path factor moves long-maturity Treasury yields more than the target factor and explains most of the contemporaneous equity response, establishing forward-looking communication as an independent transmission channel.

When the zero lower bound constrained the policy rate, forward guidance became the active instrument. For the United States, 2008–2011, Campbell et al, (2012) extract a latent guidance factor and show that guidance shocks lower intermediate-maturity yields and raise private-sector GDP forecasts, evidence that statements about the future policy path shift financial conditions and expectations even when the current rate cannot move.

Central bank announcements also reveal news about the economic outlook. If markets learn that growth or inflation is stronger than expected, long yields and equities can rise together even when policy is contractionary. Nakamura and Steinsson (2018) use co-movements of yields and stock prices to separate policy from information shocks, showing that a sizable share of event-study "policy shocks" are in fact information shocks. Jarociński and Karadi (2020) obtain the same insight for the euro area: distinguishing information from policy shocks restores the expected contractionary responses of output and prices.

The exchange rate channel

In open economies, the exchange rate responds to the relative expected path of short-term interest rates domestically and abroad. A surprise contractionary monetary policy raises the expected domestic real rate and tends to appreciate the currency. A foreign contractionary policy, specifically the Federal Funds rate, which has been shown to have significant spillovers, has the opposite effect on the bilateral exchange rate. With sticky prices, the nominal exchange rate adjusts quickly, often "overshooting" while domestic prices stabilise with lags, this makes the exchange rate a fast channel for both foreign and domestic policy (Gali and Monacelli, 2005).

A number of event study studies show that US monetary policy surprises move foreign exchange rates and foreign equity indexes contemporaneously. In a cross-country study, the US Federal Open Market Committee shifted target and path surprises together with foreign equity prices and short-term prices on the day of the announcement; there is sector-specific heterogeneity to this response; however, the spillovers are systematic (Housman & Wongswan, 2011). During the zero lower bounds period, US unconventional announcements and asset purchases also reduced foreign long-term yields across advanced economies and moved exchange rates, demonstrating cross-border portfolio balance effects (Neely, 2015; Rogers, Scotty & Wright, 2014).

Foreign policy rates affect domestic equities on two margins. First, exchange-rate translation of cash flows: a foreign contractionary shock typically depreciates the domestic currency, lifting the domestic-currency value of foreign revenues and often exporters' margins, while raising the local cost of foreign inputs and compressing importers' margins, the reverse under appreciation. Industry and firm-level responses show systematic exchange rate exposure, though hedging attenuates it, creating the "exposure puzzle" (Bartram & Bodnar, 2007). Second, global discount-rate effects: foreign policy shocks shift the global price of risk and term premia, so a foreign easing lowers safe yields, makes risky assets more valuable, reduces required returns, and raises equity valuations (Miranda-Agrippino & Rey, 2020).

The credit channel

Monetary policy affects equity valuations not only through the risk-free asset structure but also by altering the cost and availability of external finance. A contractionary monetary policy stance reduces collateral values and borrower net-worth, which widens the external finance premium required by lenders and investors; at the same time, higher bank funding costs and tighter balance sheet constraints restrict the supply of credit (Kashyap & Stein, 2000). This raises the return investors require and reduces planned spending and investment; therefore, stocks fall more than expected from a contractionary monetary policy. The widely used measure of credit, excess bond premium, rises after the policy contraction and helps explain weaker activity and lower asset prices (Gertler & Karadi, 2015). Therefore, ignoring the credit channel risks misreading stock moves as pure interest rate effects, where time-varying credit costs and loan supply are causing the change.

2.3 Empirical literature on monetary policy effects on equity prices using VAR

From the research conducted by Bjornland & Leitemo (2009), the main findings from the structural VAR are that equity prices and stock markets tend to move together within the same decision window. They use a monthly data set for the US spanning from 1982 to 2002 and include consumer price inflation, real industrial production, the federal funds rate, and a broad commodity price index to neutralise the price puzzle, which is a rise in inflation observed in SVAR literature following a contractionary monetary policy shock. This is often diagnosed as a limitation of short-run restrictions and SVAR modelling itself, along with the real S&P 500 prices. A monetary policy shock is found to have no permanent impact on the level of real stock prices, robust to alternative lag specifications. A one percentage point shock in contractionary monetary policy results in an immediate nine percentage point decline in equity prices.

Following the responses of US stocks, we analyse the effect of monetary policy through a cross-country comparison based on the findings of Neri (2002). The policy equation is constructed to respond immediately to exchange rate inflation and economic activity. The stock price equation remains unrestricted, allowing policy and equities to move together on announcement days. Importantly, for highly open economies, the nominal exchange rate is included. The analysis is conducted monthly from 1985 to 2000, with lags selected via residual diagnostics for each country. Results from the accumulated impulse response function indicate that a contractionary monetary policy has a negative and temporary impact on stocks across all considered countries, with different magnitudes and timings. Additionally, policy tightening is followed by a contraction in money supply, output, and prices. These findings are further confirmed by a DSGE model, which shows the same qualitative decline in equity prices.

One of the most frequently cited papers on VAR modelling of monetary policy on equity prices is Thorbecke (1997), which provides a monthly account that includes common macroeconomic controls and returns on 22 CSRP industry portfolios. Monetary shocks are identified through orthogonalised innovations in the funds rate; a narrative measure of policy is also considered to test if exposure to policy is priced, along with an event study for more detailed analysis. All mentioned approaches suggest that the effect of expansionary monetary policy on stocks remains positive, with stronger effects in cyclical and financially sensitive industries, indicating country-wise heterogeneity as seen in previous literature. Thorbecke (1997) argues that countries differ in their exposure because surprises affect both expected future cash flows and the discount rate used by investors. Therefore, assets that react more to these shocks experience larger price swings, and the variation between countries arises due to differences in financial structures, policy rules, or economic openness.

To directly assess how stock reactions differ between a small open economy and a larger one, Li, İşcan, and Xu (2010) estimate two SVARs for Canada and the United States. These are identified with contemporaneous restrictions based on central bank reaction functions and market-clearing conditions and are augmented with portfolio-shock innovations to capture time-varying equity premia. Their primary finding is that a contractionary monetary policy shock reduces equity prices in both countries, but the impact and persistence are significantly greater in the United States. This asymmetry aligns with Canada's openness and the fact that a substantial portion of Canadian financial conditions are effectively imported via U.S. policy stance and the exchange rate, so domestic shocks are diluted compared to the U.S. The comparison provides a strong foundation for understanding how spillovers can influence domestic monetary policy through cash-flow channels and highlights the importance of considering foreign interest rate conditions in studies of domestic shocks, particularly in open economies.

To further demonstrate the persistence of foreign spillovers and the significance of cross-country comparison, we examine the cross-country open economy identification by Kim and Roubini (2000). They utilise a non-recursive VAR and monthly data that includes the policy rate, money supply, inflation, and output. Policy is influenced contemporaneously by the exchange rate. Their specification resolves the classic price and liquidity puzzle by incorporating the exchange rate. Following a contractionary monetary policy shock, the exchange rate increases on impact and gradually depreciates, while equity prices fall. These results and variable choices are crucial for cross-country comparison, as controlling for foreign economic activity helps isolate domestic shocks and ensures the identification remains economically coherent.

However, we find that more recent literature on monetary policy transmissions has focused on model specifications or identifying high-frequency shocks. The recent literature on the Euro region concentrates more on identifying high-frequency domestic shocks using sign restrictions in a Bayesian VAR, employing the Euro STOXX, S&P 500, USD, and EUR exchange rates. These studies find that European policy moments are dependent on foreign announcements and are reflected in daily co-movements and announcements (Brandt et al., 2021). The study indicates that US shocks account for nearly forty percent of the variation in Euro region long-term yields and equity prices on average, while domestic policy continues to exert a negative impact on equities.

These findings are echoed by further research on national Euro region stock markets, which observe heterogeneous effects stemming from foreign announcements and shocks. Nave & Ruiz (2018) demonstrate that controlling for foreign policy and domestic responses reveals heterogeneous effects on equities, although they still exhibit the expected decline following contractionary monetary policy. They highlight that the transmission varies depending on listed firm characteristics and the extent to which each country's implied Taylor rule stance deviates from the ECB's overall stance. This serves as a relevant example illustrating cross-country equity differences within a monetary union, and controlling for foreign conditions enhances the clarity of identification.

2.4 Summary of literature and scope of research

The literature on monetary policy and theoretical transmission channels offers a foundation for cross-country comparisons. With forward-looking agents, systematic rules do not buy a permanent inflation and unemployment trade-off; real effects arise when policy surprises revise the expected path of rates (Phelps, 1967; Lucas, 1976; Barro & Gordon, 1983; Clarida, Galí & Gertler, 1999). In asset pricing, the present-value identity implies that monetary policy moves equities mainly by altering required returns rather than instantaneously changing dividends; variance decompositions and announcement-window evidence confirm the dominance of discount-rate news (Campbell & Ammer, 1993; Bernanke & Kuttner, 2005). In small open economies, exchange-rate and foreign-rate channels transmit policy to domestic discount rates and cash flows, so domestic identification that ignores foreign conditions risks misattributing spillovers to domestic shocks (Kim & Roubini, 2000).

Crucially, many influential studies model open economies, omitting exchange rate and foreign-policy blocks either by design, accounting for world economic activity or data limits Chatziantoniou, Duffy and Filis, 2013). By contrast, high-frequency event-study and foreign rate-inclusive models indicate that foreign policy is a first-order factor for local equities: U.S. FOMC surprises immediately impact the FTSE (Wongswan, 2009), and lower Norwegian stocks follow U.S. tightening (Ehrmann & Fratzscher, 2006). Consistent open economy evidence suggests foreign rates explain non-trivial domestic stock variance, and joint domestic-foreign identification sharpens the home estimate (Li, İşcan & Xu, 2010; Ha, 2023; Alessi & Kerssenfischer, 2016; Jarociński and Karadi, 2020).

In summary, our study addresses these gaps via a UK and Norway comparison of two inflation-targeting, highly open economies with heterogeneous market structures while controlling for foreign policy rates to isolate the domestic equity response.

Chapter 3: Methodology

3.1 Research approach and framework

We estimate two models, a recursive (Cholesky) SVAR and a short-run-restricted SVAR with an explicit foreign-spillover block, building on Chapter 2. Identification follows policy theory and decision windows; stability is checked across models. First, a recursive SVAR excluding the foreign rate recovers conventional stock responses and validates the variable order (Bjornland, 2009). Second, an open-economy short-run-restricted model, as suggested by Li, İşcan, and Xu (2010), incorporates foreign-rate spillovers, with zero restrictions supported in the literature (Chatziantoniou, Duffy & Filis, 2013). We report 24-month accumulated IRFs and structural FEVDs to gauge policy's contribution to equities and use local projections of accumulated responses for robustness (Jordà, 2005), reducing sensitivity to lag choice and VAR-propagation assumptions.

3.2 Selection of countries and justification

To evaluate the impact of monetary policy across two advanced economies that serve as a comparative basis, we select the UK and Norway. These economies operate under transparent, rule-based, inflation-targeting regimes with independent central banks and floating exchange rates, conditions under which modern open economy monetary theory offers predictions for asset price transmissions.

The Bank of England's monetary policy committee gained operational independence in 1997 with systemic inflation targeting rules (Geraatas, 2024). Norway adopted flexible inflation targeting shortly after, in 2001, with a revised regulation in 2018 setting a two percent operational target (Roisland, 2017). Similar mandates and communication frameworks are important, as they ensure policy surprises are interpreted consistently across countries, an essential precondition when employing high-frequency identification and SVARs to trace the equity channel (Gürkaynak, Sack, & Swanson, 2005).

Despite institutional comparability, the two economies differ along lines where theoretical comparisons can provide valuable information. For example, Norway is a highly open small economy with high trade exposure; the exchange rate is a first-order transmission margin in sticky-price open economy models, and empirical evidence suggests that a contractionary monetary policy shock appreciates the exchange rate contemporaneously (Kearns & Manners, 2006). The pound is also freely determined by foreign exchange markets; however, the UK's larger domestic market and capital-dependent market implies a distinguishable mix between discount rate and cash flow channels between countries, therefore, a comparison can be effective and identify the monetary policy channels of transmission.

Sectoral composition also plays a significant role in listed equity, Norways market is energy reliant as of 2025 the Norway index allocates a large share to energy and financial sectors moving stock valuations mechanically more sensitive to oil and exchange rate related cash flows compared to broader UK markets with Financials, Consumer Staples and Industries taking a larger share, which tilts the market towards globally diversified cash flows, whose immediate sensitivity to domestic oil shocks and exchange rate is weaker

Financial transmissions also differ across the UK and Norway as their mortgage contracts weigh differently on the cash-flow channel. In Norway, adjustable-rate mortgages (ARMs) are more prevalent; therefore, policy rate changes reprice monthly payments swiftly. This feature has been identified in cross-country evidence as amplifying monetary transmission to housing and consumption. When ARMs are prevalent, the response of residential investment and house prices is stronger, and consumption is higher (Calza, Monacelli, and Stracca, 2013). By contrast, the UK mortgage market is fixed-rate dominant but at short maturities, which shows pass-through and bunches it at refit dates; UK household-level evidence shows the aggregate consumption response to rate changes is driven by mortgages (Cloynee, Ferreira and Surico, 2020).

3.3 SVAR modelling

We choose the SVAR model for this analysis for economically motivated and interpretable shocks and their overall indirect effects from multivariate time-series data. Relative to a reduced form VAR, which only traces correlations among residuals, the structural model augments a small set of economically motivated restrictions, like our short-run zero restrictions, to map residuals into monetary policy impacts, so that impulse responses, forecast error variance and historical decompositions have causal interpretations (Kilian & Lütkepohl, 2017). This provides a middle ground compared to the strong parametric and equilibrium restrictions imposed by DSGE models and is more informative than event study approaches, as we get the simultaneous equations for economic activity, allowing contemporaneous feedback within announcement windows (Bjornland & Leitemo, 2009).

We begin our modelling with the recursively identified VAR via a Cholesky decomposition of the reduced error covariance matrix, which imposes a lower triangular contemporaneous impact matrix and thus a causal ordering. Variables earlier in the ordering can move the later contemporaneously within the month, but not vice versa.

Recursive Model

Ordering

$$y_t = [\ \text{WIP},\ \text{IPI},\ \text{CPI},\ \text{M3},\ \text{POLICY},\ \text{EXRATE},\ \text{STOCKS}\ ]$$

$$A_0 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_{21} & 1 & 0 & 0 & 0 & 0 & 0 \\ a_{31} & a_{32} & 1 & 0 & 0 & 0 & 0 \\ a_{41} & a_{42} & a_{43} & 1 & 0 & 0 & 0 \\ a_{51} & a_{52} & a_{53} & a_{54} & 1 & 0 & 0 \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & 1 & 0 \\ a_{71} & a_{72} & a_{73} & a_{74} & a_{75} & a_{76} & 1 \end{bmatrix}$$

World economic activity is characterised by a sizeable body of empirical literature that acknowledges a small set of global real activity factors explains a substantial share of the business-cycle movement across advanced economies, while country-specific shocks account for a much smaller and slower-moving component of activity. In the study of G-7 countries, Kose, Otrok, and Whiteman (2003) found that a global output factor dominates national fluctuations, implying that shocks to a single economy are part of the global signal at both monthly and quarterly horizons. In SVAR literature, Pesaran, Schuermann and Whiteman (2004) implement this by treating foreign aggregates as weakly exogenous for individual economies. Findings show that trade-weighted foreign variables can, in practice, be considered exogenous, as feedback from a single economy is quantitatively limited within the monthly window.

Extending the approach to Europe, Dees et al. (2007) find that weak exogeneity is not rejected for the majority of the countries in the Euro region; therefore, the global driver can be treated as block exogenous in SVARs without the loss of empirical information. In the UK context, Mumtaz and Surico (2009) use a VAR model and document that international factors explain a large fraction of UK macro variation, consistent with the view that the UK mostly receives global shocks rather than shifting the global factor on impact. For Norway, Aastveit, Bjørnland and Thorsrud (2016**)** show in a multi-country factor model that small open economies are primarily driven by common global economic activity forces, with domestic shocks playing a much smaller role in explaining co-movement at business-cycle frequencies, and additionally, the case of oil trade plays a significant role in the Norwegian economy.

The domestic output comes second in the matrix, only affected contemporaneously by global economic activity. Production, hiring and capacity utilisation adjust with frictions; therefore, a monetary policy surprise should not move real output contemporaneously, while a global demand shock can show its effects on output through export-dependent sectors. Recent literature shows that output responses to policy are delayed, peaking only beyond one quarter, which is the New-Keynesian prediction under sticky prices and adjustment costs.

For the UK, evidence suggests that unemployment rises and output falls after a tightening; however, there is little evidence of a within-month output move, which justifies ordering the output before the policy instrument at the monthly frequency (Cloyne & Hurtgen, 2016). For Norway, macroeconomic structural evidence shows that transmission to activity builds gradually, such as household-level pass-through and slow aggregate demand effects (Holm, Paul & Tischbirek, 2021).

By contrast, global activity, a dominant global component in production and trade, can transmit effectively via orders, contemporaneous input and output links, especially in Europe's open economies. (Huo, Levchenko and Pandalai, 2019) shows that shocks propagate contemporaneously across countries through GDP co-movements, the production block can reduce this effect so that policy shocks are not contaminated by global demand news.

Prices are contemporaneously tied to global conditions and weakly to domestic real activity, while remaining predetermined to the policy instrument at the monthly frequency. Cross-country evidence shows a strong global component in inflation that moves at the monthly horizons (Ascari & Fosso, 2024). They show that international forces increasingly drive the cyclical part of inflation, with slow-moving cost-push trends determining the magnitude of the effect. Complimenting this, Dao et al. (2024) decompose the inflation cycles across 21 countries and find that headline shocks and their pass-through into core inflation explain more of the fluctuations.

For European countries, high-frequency identification evidence shows no impact on prices from the contractionary monetary policy with immediate effect. CPI falls with multi-quarter lags after a contractionary shock rather than on impact (Caesar-Bianchi, Thwaites and Vicondoa, 2020). For open economy SVARs, document gradual price responses to monetary tightening due to sticky prices and adjustment frictions (Bjornland & Jacobsen, 2010). Therefore, the ordering keeps CPI isolated from instantaneous policy moves while letting the global economic activity explain the contemporaneous price news that would otherwise impact the policy shock.

The money supply is an observable state variable that captures money demand shocks and portfolio composition and shifts that are not policy shocks, stabilises nominal dynamics at business-cycle horizons and can help improve identification in extended period monthly VARs (Binner et al., 2025). The model recovers sensible impulse responses once a monetary aggregate has been included. Cross-country evidence also shows stable money demand relationships, informing the use of broad money as an informative state, even under inflation targeting and broad money growth to nominal spending and inflation in the post-2020 cycle, argues that money aggregates carried timely information that standard rate only models omit (Barnett et al., 2023)

The policy rate pass-through is also restricted, as deposit rate pass-through is incomplete and sluggish; large cross-country comparisons find heterogeneous and partial pass-through to deposit rates (Beyer et al., 2024). However, we have considered the channel pass-through in the open economy model discussed later in the methodology.

At the monthly frequency, the macro and nominal variables are slower-moving relative to the surprise component of policy. A UK-focused study shows the contractionary policy effect reduces output and lowers inflation after approximately two to three years (Clyone & Hurgten, 2016). In contrast, exchange rate and equity prices are comparatively fast-moving variables as they reprice contemporaneously to the policy news. However, in the diagnostic model specification, monetary policy is affected by CPI, output and global activity, which could conflate the direct effect on stocks. At the monthly frequency, the policy surprise is defined relative to expectations of central bank meetings, while the later revisions of prices or output are not yet released or already in forecasts; therefore, same-month movements in this series should not co-determine the stocks. This concern is addressed in the baseline model.

Lastly, we have the equity prices and exchange rate that reprice within short windows of unexpected policy announcements. By arranging them last, we only get the pass-through feedback from the slow-moving macro variables. The Efficient Market Hypothesis (Fama, 1970) supports this ordering, as publicly available news should be reflected in asset prices immediately. Therefore, any unanticipated policy surprise should be reflected in financial prices within minutes, rather than over months or quarters.

Following the ordering of the variables in our model, we now discuss the inclusion of a foreign policy rate, specifically the US federal funds rate, the choice of the federal funds rate as the control for foreign policy, discussed later in the data selection section. We utilise the short-run restrictions proposed by (Li, İşcan and Xu, 2010).

Baseline Model

$[R_t^*, Y_t, P_t, M_t, R_t, E_t, S_t]$ = [foreign rate, output (IPI), prices (CPI), money (M3), policy rate, exchange rate, stock prices].

Structural shocks (row mapping): $\varepsilon_t = [\varepsilon_t^{FR}, \varepsilon_t^{Y}, \varepsilon_t^{P}, \varepsilon_t^{M}, \varepsilon_t^{R}, \varepsilon_t^{E}, \varepsilon_t^{S}]$

$$A_0 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_{21} & 1 & 0 & 0 & 0 & 0 & 0 \\ a_{31} & a_{32} & 1 & 0 & 0 & a_{36} & 0 \\ 0 & a_{42} & a_{43} & 1 & a_{45} & 0 & 0 \\ 0 & 0 & 0 & a_{54} & 1 & a_{56} & 0 \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & 1 & 0 \\ a_{71} & a_{72} & a_{73} & a_{74} & a_{75} & a_{76} & 1 \end{bmatrix}$$

The US rate is contemporaneously exogenous to the domestic block. There is sufficient evidence to show that US monetary policy surprises propagate contemporaneously and globally through the discount rates, risk premiums and cross-border credit as a part of the "global financial cycle, therefore, it can impact the UK and Norway financial conditions before the domestic decisions (Miranda-Agrippino & Rey, 2020). In cross-country comparison evidence, a surprise US contractionary depreciates foreign currencies against the dollar and lowers domestic output and inflation, confirming that US policy shocks are the first-order external driver for small open economies (Dedola, Rivolta & Stracca, 2017).

Output is affected contemporaneously by foreign shocks as external monetary surprises can quickly affect orders and financial conditions, as also shown by (Dedola, Rivolta & Stracca, 2017). The UK's globally intermediating financial system makes sterling credit conditions and USD funding sensitive to US rate shocks that transmit to firms' working capital and trade finance immediately. Norway's oil-dependent supply chain is highly sensitive to exchange rate and commodity fluctuations. Therefore, as US policy tightening often coincides with stronger USD and lower USD commodity prices, and the Krone depreciates, it can cause early setbacks in petroleum-related services and goods production in the same month.

Prices at the monthly horizon are contemporaneously affected by foreign rates, domestic output and the exchange rate. Prices affected by the exchange rate are justified by the literature on the exchange rate pass-through into consumer prices, which is state-dependent and much higher when driven by monetary and financial shocks, as in the case of the UK (Forbes, Hjortsoe, & Nenova, 2018). Pricing in dominant currencies and staggered resets imply that import price changes transmit to CPI more rapidly after monetary shocks (Gopinath, Itskhoki, & Rigobon, 2010).

Broad money reflects banks' balance-sheet pricing and households' or firms' portfolio choices that reprice when the policy rate moves via deposit-rate setting, deposit spreads, and the mix of transaction vs time deposits. This delivers a contemporaneous policy effect through the deposit channel (Drechsler, Savov & Schnabl, 2017). In our setting, retail deposits are rate-elastic and policy-sensitive; when stance changes, banks' deposit pricing and household demand co-move, so money responds within the month. Li, İşcan & Xu (2010) estimate monthly VARs in log levels (all variables in logs except interest rates) and, in the policy rule, condition on contemporaneous M2 with lags of all variables, so money loads both on impact and through prior-month lags. By contrast, we model M3 in monthly log differences, so we do not add a separate predetermined (lagged) money term, while still capturing within-month pass-through.

The external policy shock moves asset prices and yields abroad contemporaneously, but for resident broad money, the immediate transmission operates through local deposit pricing, which is governed by domestic pass-through from the home policy rate and by banks' funding structures; cross-border monetary news does not reprice UK and Norwegian retail deposits or reallocate household deposit quantities within the month in a systematic way. Large-scale comparative studies for Europe document that pass-through to deposit rates is incomplete and gradual, with material adjustment frictions, so contemporaneous foreign-rate effects on domestic deposits are weak relative to the dominant domestic channel (Beyer et al., 2024)

The domestic policy rate is affected contemporaneously by the exchange rate, a feature that mirrors actual price inflation in targeting small open economies, where central banks observe the exchange rate while setting interest rates, improving the model fit for our countries (Lubik & Schorfheide, 2007). By contrast, as mentioned before, the same monthly realisations of prices and output should not load into the policy surprise. In both the UK and Norway, the central bank targets a short rate and uses open market operations to steer it. Unexpected shifts in liquidity demand and credit creation require immediate adjustments in the policy stance or the path-forward guidance to keep the overnight rates aligned with the target. This gives feedback from money and credit to the policy instrument, even though the medium-run policy movements impact money supply.

We also restrict the contemporaneous effect of foreign policy on domestic monetary policy as, in monthly samples, policy instruments adjust infrequently with inertia; therefore, foreign rate effects work through fast prices like yields and exchange rates and affect policy via lags. A common assumption in inertial Taylor rule estimates and interest rate smoothing, where the central bank deliberately moves gradually (Rudebusch, 2002)

The fast-moving variables are ordered last and maintain the same restrictions as the default model; however, here we allowed both to be affected by the foreign monetary policy conditions to assess the effect of including foreign rates in small open economies.

Additionally, we include WIP exogenously in our model and see negligible difference in the variance contributed to monetary policy on predicting stocks and the LR test indicates omitting WIP results in better model fit therefore, following the Li, Iscan and Xu, (2010) convention where they test for the difference in explanatory power including oil prices and omission based on minimal difference, our baseline matrix and results are reported excluding global economic activity, discussed in section 6.2.

Chapter 4: Data and descriptive statistics

In the following chapter, we discuss our choice of variables and justification for the variable repressing the macroeconomic activity we want to account for. Our data sample is monthly and extends from January 1997 to December 2023 and includes 324 observations. The detailed data reference code, methodology and sources can be found in Table 3 in the appendix for ease of replication. We choose to use monthly data to match the frequency of the macroeconomic variables considered in this study. The data set starts in 1997, accounting for the central bank independence of the UK, followed closely by Norway in 2001.

4.1 Data sources and sample description

World Industrial Production (WIP)

For a monthly open economy SVAR, we use the WIP as the external real activity block because, at the monthly frequency, it is a more coherent and predictive measure of global real activity compared to an oil market proxy or shipping-based global activity. Hamilton (2021) audits the construction of the shipping costs index and finds that movements in freight rates are contaminated by shipping-supply shocks, which are unrelated to the global business cycle. The paper treats indicators on three criteria: trend treatment, coherence with world output and the ability to predict world commodity prices. On all three measures, WIP performs better, which tracks the world business cycle more closely.

As mentioned in the literature, Norwegian equities are significantly affected by oil prices; therefore, we use Brent Oil prices as a robustness check to verify that our results are not an artefact of misattributed shocks discussed in section 6.2. For our baseline and recursive specification, we use the WIP as the control, as oil markets embed multiple structural shocks; therefore, an oil shock confounds global activity with shocks unrelated to global demand (Killian, 2009).

Industrial Production Index excluding construction (IPI)

Using IPI for our model has several reasons. At the measurement level, the production indices we want to load contemporaneously on external conditions are output-based aggregates. By excluding construction, we want to transmit trade-driven shocks; the output measure should be for tradable-based industries and not a hybrid that mixes labour input for site-based construction. It moves contemporaneously with the global cycle and therefore transmits the external conditions onto the domestic block appropriately without noise. The international business cycle literature establishes a common world factor that explains a large share of the variance in macro aggregates; across countries, industrial production remains the preferred monthly proxy (Kose et al., 2003).

Consumer Price Index Headline (CPI)

Within the CPI classifications, the headline or total CPI is preferred here over the trimmed or core CPI, as it allows for the identification of exchange rate pass-through channels contemporaneously in our baseline model. Empirically, the exchange rate pass-through to consumer prices is shock-dependent and significantly stronger when monetary and financial shocks drive exchange rate movements; therefore, excluding energy and food prices would attenuate the intended channel (Forbes, Hijortsoe and Nenova, 2018). Literature additionally supports our choice under dominant currency pricing and sticky local prices; the composition and timing of passthrough into prices are captured by headline CPI, but core indices may mute this effect (Gopinath et al., 2020). Therefore, for effective comparability and application in our baseline model, we use CPI headline instead of core CPI.

Broad Money supply (M3)

Following the convention used by Li, Iscan & Xu (2010), we include the broad money component in our analysis. However, we use the M3 aggregates instead of M2, as the Norwegian M2 series is not comparable across the 2015 redesign of the monetary aggregate statistics. Using M3, therefore, avoids a break in the series. Economically, M3 encompasses marketable money-like instruments that are relevant for asset pricing and bank funding. This matches the euro-area SVAR in Cassola & Morana (2004), who include real M3 and real stock prices in the transmission block when tracing policy shocks to equities. In our implementation, we work with monthly log differences of real M3, keeping definitions harmonised across the UK and Norway while preserving the asset-pricing-relevant liquidity margin.

Central Bank Rates (Policy Rate)

We use the central bank policy rate as it stands as the real instrument set at scheduled meetings, and in modern floor systems, it pins the overnight money market rates. Banks' reserves are remunerated at the policy rate; therefore, overnight rates do not lie far from it. For a monthly SAVR, this provides a clean decision-based measure of stance that maps directly onto the reaction function we want to identify. We choose not to use overnight rates such as SONIA for the UK and NOWA for Norway, as they are market outcomes that can deviate from the stance due to liquidity conditions; they are more appropriately used to measure announcement window surprises and not instruments for setting interest itself.

Nominal Effective Exchange Rate (NEER)

We choose the NEER as our exchange rate measure for both countries; it is a geometric weighted average of collective bilateral exchange rates, therefore capturing price competitiveness and monetary transmission. For the UK and Norway, NEER is the official operational measure used by the central banks; for Norway, this is especially important as a single bilateral exchange rate would not reflect the diversified trade basket. Additionally, as NEER is a trade-weighted index, the sign maps synchronously to currency strength against the entire basket; therefore, a positive accumulated response function indicates a broad-based appreciation, which avoids the ambiguity found in interpreting comparative currencies.

Equity prices

We utilise the headline equity indices, the FTSE 100 for the UK and the OBX index for Norway, as they are the most liquid and policy-sensitive measures of domestic equity valuations. Both are large-cap, free-float weighted benchmarks that reprice immediately to monetary policy surprises via the discount rate channel, ideal for monthly identification. The FTSE 100 also captures the UK's multinational exposure; therefore, the effects of policy and exchange rate can also be studied. To ensure the results are not an artefact of index composition, as sectors can show severe heterogeneity, we also include the OECD financial market share price, which ensures a harmonised cross-country comparable benchmark.

US federal funds rate (Foreign Rate)

We take the US federal funds rate as the foreign policy block, as the US monetary policy block dominates the global spillovers. By contrast, using a single European policy rate, e.g., the ECB rate, as the foreign block would understate the external driver most likely to contaminate domestic shocks in the UK and Norway. High-frequency measures show ECB surprises primarily reprice euro-area assets, with far more limited same-day effects outside the block compared with FOMC shocks (Altavilla, Giannone and Lenza, 2019). Structurally, the dollar's dominance in trade invoicing and global intermediation means U.S. policy transmits into import prices, term premium and global risk-taking even when bilateral EUR links matter locally; a single foreign rate variable is therefore best assigned to the Federal funds rate to absorb the broad, system-wide discount-rate and funding shocks.

4.2 Variable transformations

We choose to use real equities to ensure movements reflect discount rate news and real cash flow news, and not inflation. In present value terms, valuations are real payoffs discounted by an SDF. As we use a price-only index, the log change in real price is a close proxy for real capital gains.

Differencing turns a non-stationary level series into a stationary growth rate by removing the common stochastic trend. Taking log differences yields approximate percentage changes with stable variance; therefore, the VAR can model without spurious regressions. The loss is long-run cointegration not being modelled, but it fits the purpose of identification of short-run dynamics. The policy rate is an additive control variable; shocks are interpreted as level changes in percentage points. Therefore, modelling interest rates in levels keeps IRFs interpretable; a log transformation would cause proportional changes and breaks near zero.

$$\Delta \ln X_t = \ln X_t - \ln X_{t-1} \approx \% \text{ change per month}$$

4.3 Validity tests

A Structural VAR assumes the reduced form is covariance-stationary, so that shocks have a well-defined moving average representation and impulse responses converge . To validate this assumption, we have transformed the data as mentioned above and test for unit roots using the Augmented Dickey Fuller test across the transformed series, ADF rejects a unit root null hypothesis at the 1% level (Appendix Table A3 A4), indicating that VAR inputs are stationary in differences, this allows us to estimate a VAR in differences rather than a VECM as we are not modelling long-run cointegration and our focus is the short-run dynamics.

We select the lag order with a standard triangulation. The information criteria favour short-run dynamics. AIC points to 2 lags for the UK and 3 for Norway, whereas the LR test supports a longer specification at lag 6. We set the baseline lag to 6, as this follows the convention used by Li, Iscan & Xu (2010), which is sufficient to capture quarterly propagation and announcement cycles while being parsimonious enough to preserve degrees of freedom and avoid overfitting. At lag 6 we find very mild autocorrelation at the first month that does not persist to higher orders. As an additional specification, we also report the effect of monetary policy on stocks spanning a full calendar year, allowing for annual mean reversion (Peersman & Smets, 2003).. This analysis reveals that prices return to zero, in line with the theory of money neutrality, and all roots are stable for 12 lags. For our baseline model, a 7-variable SVAR requires 21 contemporaneous zero restrictions; we have imposed 22, therefore the model is over-identified by one restriction, which is not rejected by the likelihood-ratio test at the 5% level (Appendix Table A9, A10).

4.4 Descriptive statistics

Table 1: Descriptive statistics for transformed variables UK

	N	Mean	SD	Min	p25	Median	p75	Max	Skewns	Kurtoss
dIPI	323	.0005	0.0171	-.1661	-.0056	.001	.0071	.1007	-2.2292	35.9885
dCPI	323	.002	0.0033	-.007	0	.0024	.0038	.0212	.4428	7.231
dM3	323	.0045	0.0085	-.0405	0	.0044	.0091	.0791	1.6319	22.4517
dPolicy	323	-.0021	0.1799	-1.5	0	0	0	.75	-2.3151	22.3831
dEX	323	-.0002	0.0149	-.0676	-.008	.0002	.0096	.0421	-.8724	5.7747
dFTSE	323	-.0001	0.0401	-.1486	-.0212	.0046	.0266	.1174	-.6574	4.036
dOECD	323	-.0001	0.0372	-.2419	-.0161	.004	.0215	.0985	-1.6148	10.5502
dFR	323	.0004	0.2177	-1.5	0	0	0	.75	-2.1093	17.4167

Table 2: Descriptive statistics for transformed variables Norway

	N	Mean	SD	Min	p25	Median	p75	Max	Skewns	Kurtos
dIPIN	323	-.0004	0.0295	-.1166	-.0167	0	.0146	.1128	.0292	5.259
dCPIN	323	.002	0.0046	-.0128	-.001	.0014	.0046	.0235	.365	4.6031
dM3N	323	.0053	0.0097	-.0598	.0007	.0052	.0102	.0454	-1.368	12.9432
dPOLICYN	323	-.0002	0.2236	-1.14	0	0	.01	1.93	1.3713	24.4924
dEXN	323	-.0009	0.0159	-.0893	-.0091	-.0008	.0093	.0459	-.7837	6.278
dOBX	323	.0052	0.0591	-.307	-.0227	.0103	.0422	.1567	-1.2937	7.4062
dOECDN	323	.0054	0.0561	-.3291	-.021	.0118	.0386	.1933	-1.4199	9.0743
dFR	323	.0004	0.2177	-1.5	0	0	0	.75	-2.1093	17.4167

Preliminary Observations

In the UK, industrial production growth is flat on average (0.05%) but prone to large contractions (min -16.6%). Headline CPI changes are small and smooth (mean 0.2%). Broad money supply sees modest growth at the monthly horizon (0.45%) with wide tails, consistent with occasional liquidity surges. Policy changes are infrequent, and the negative mean reflects net easing over the sample, asymmetric adjustment during a crisis. Exchange rates show moderate dispersion, while stocks are volatile as expected.

In Norway, the industrial production contracts slightly on average (-0.04%) and is comparatively more volatile. Policy moves to a larger and more two-sided matching regime where rises and cuts both arrive in sizable steps. Exchange rate changes are modest on average but more volatile, and equity returns are higher and riskier (SD 0.22, min -30.7%), consistent with energy and commodity exposure and exchange rate-sensitive cash flows.

The rolling-window means show common breaks in both countries, equity and output drops are most significant during FC and pandemic period, CPI is broadly stable until the post-2021 surge, and policy at the lower bounds is lifted in 2022-23, and step moves in exchange rates. These patterns suggest treating exchange rates and equities as fast-moving variables, and IPI and CPI are slower-moving variables.

The correlation panels are consistent with that timing. Policy and EX are mostly negatively correlated in the UK and more variable in Norway, motivating contemporaneous EX responses. Correlations of policy with equities are close to zero and state-dependent, which argues against including equities in the policy rule. Correlations of policy with IPI and CPI are weak, supporting their predetermined status. Correlations of policy with M3 tend to be negative after lift-offs, consistent with same-month pass-through from policy to deposit pricing, not the reverse. These are descriptive patterns (not causal) and 30-month windows smooth short-lived reversals, but they align with the short-run restrictions used in the SVAR.

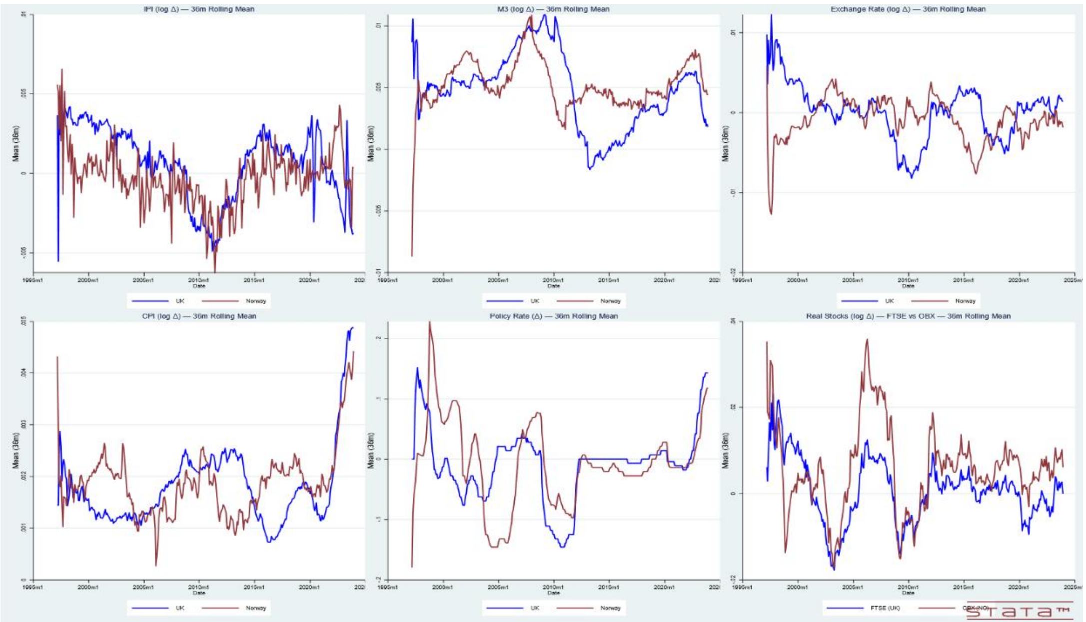

Figure 1: 36-month rolling plots

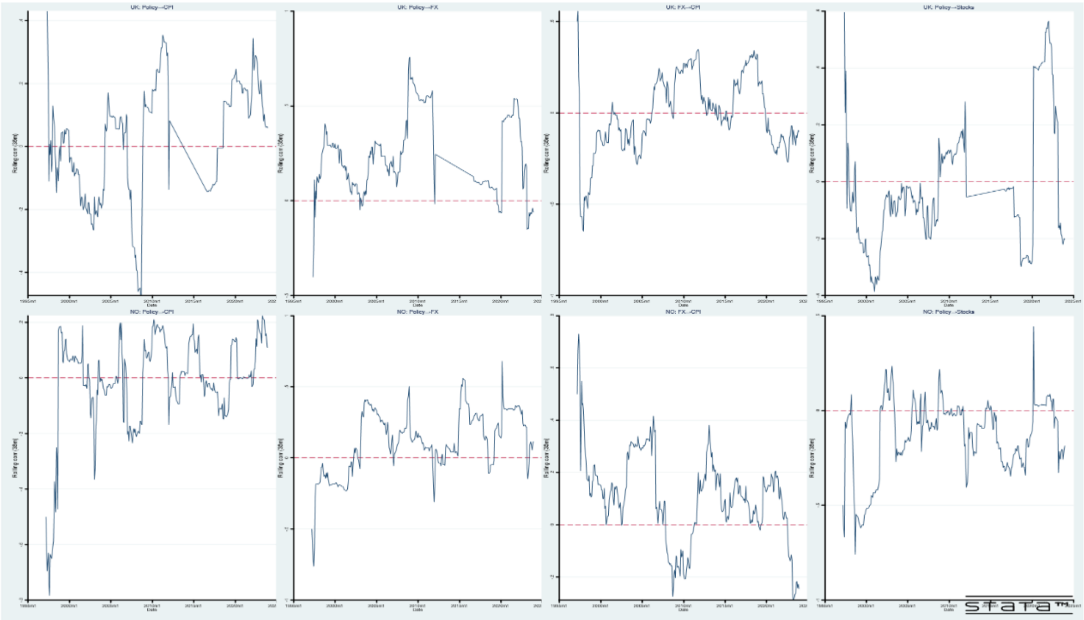

Figure 2: Rolling 36-month correlations

Chapter 5: Results

In the results section, we report the accumulated responses to a one standard deviation monetary policy innovation. We have reported both 68% and 95% confidence intervals, estimated using the Monte Carlo bootstrap simulation with 2000 repetitions, to remove imprecision in monthly samples and account for non-Gaussian errors, as suggested by Killian and Lutkepohl (2017). We interpret the results within 68% confidence intervals, as they approximate the one standard error effect, while 95% intervals provide the reader with a more comprehensive understanding of statistical significance. All impulse responses to monetary policy are interpreted as the effect following a one standard deviation rise (contraction) in monetary policy.

5.1 Model 1: Norway recursive Cholesky ordering

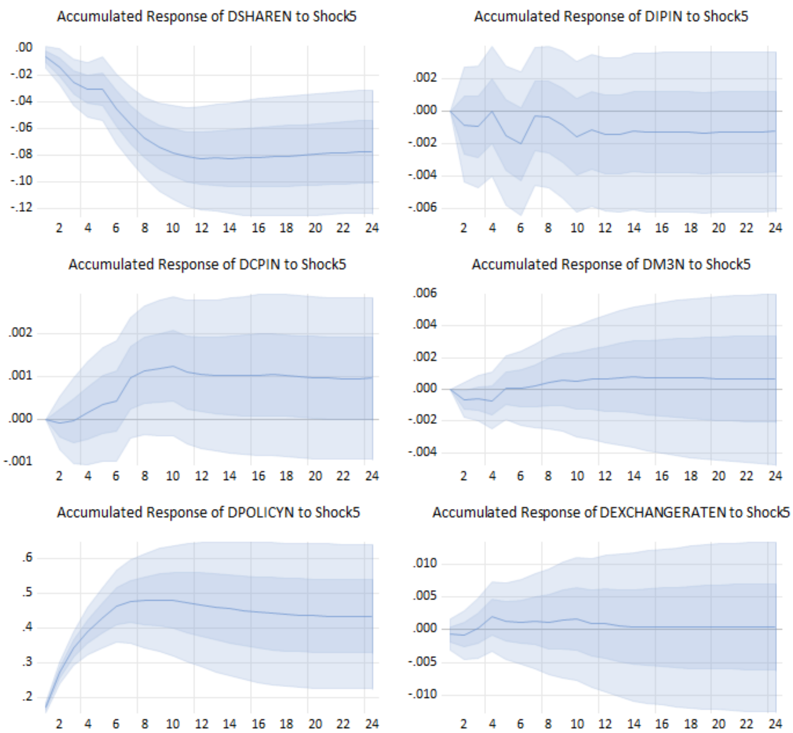

Figure 3: Impulse response functions for Norway

The result reported in Figure 1 suggests that following a one standard deviation contractionary monetary policy shock, the real equity level path reacts immediately, as the median response is below zero in the first month, and at the 68% confidence level, and shows a fall in real equity prices. This result, although with relatively high magnitude due to a lack of scaling, aligns with the findings from the literature review. Under the efficient market hypothesis, only the surprise component of the policy announcement should move equities, and it should do so immediately (Fama, 1970).

Additionally, the negative impact aligns with the cross-country evidence on small open economies in the Euro region, which finds a systematic contemporaneous drop in equity returns after rate hikes (Ioannidis and Kontonikas, 2008). The immediate onset and persistence align with intermediate asset pricing theory, where contractionary policy can compress intermediate capital, raising the price of risk (He and Krishnamurthy, 2013).

Moving to Industrial production, the median drifts negatively; however, the results are insignificant at the 68% level. Although consistent with the theory that a contractionary monetary policy should have a negative impact on output as borrowing costs rise, followed by a reduction in investment, the insignificance of this effect suggests that at the monthly frequency, production plans and hiring adjust with friction, policy shocks propagate to real activity with a delayed effect. Transmission channel literature suggests that output responses typically react after one quarter following the New-Keynesian inter-temporal substitution with sticky prices (Boivin, Kiley and Mishkin, 2010).

We see the classic price puzzle in our CPI response to a contractionary monetary policy, although there is no significance of this effect at the 95% confidence interval. The positive impact is explained by the theoretical interpretation of central bank raising interest rates in response to private information on future inflation or omitted cost-push shocks, if these drivers are not modelled explicitly in the model, the identified policy shocks inherits inflation-pressure news therefore, this is a result of omitted variable bias rather than the claim that policy causes inflation to rise (Sims, 1992)

The exchange rate appreciates following a contractionary monetary policy shock as expected from theoretical literature, although insignificant at both the confidence interval levels, the appreciation is consistent with a tightening that raises the expected short rate path and with stock prices appreciating the currency over the window (TerEllen, Jansen and Midthjell, 2020).

5.2 Model 2: UK recursive Cholesky ordering

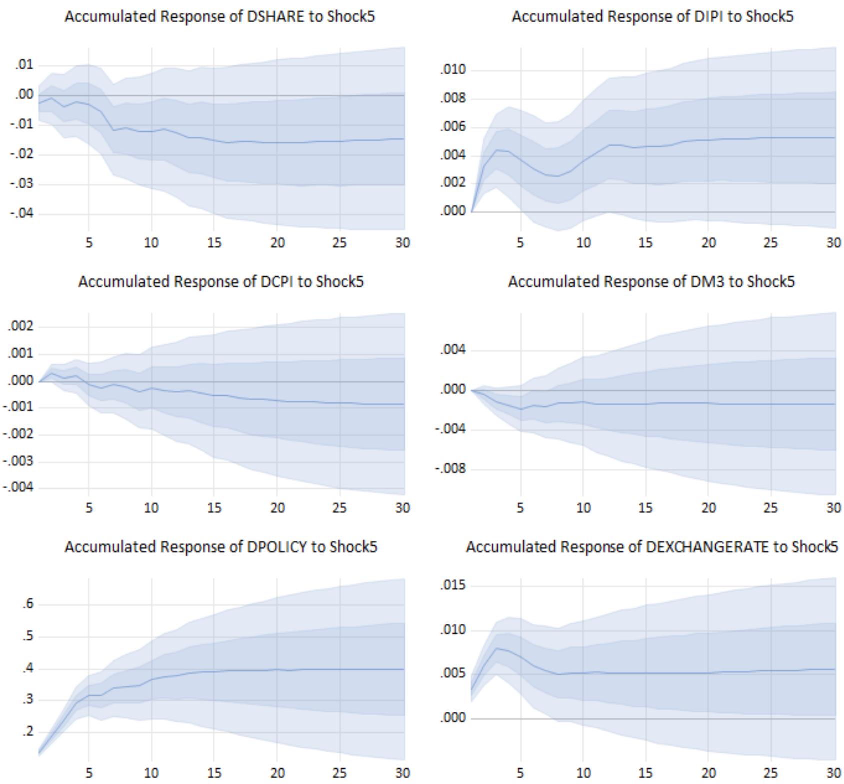

Figure 4: Impulse response functions for the UK

From Figure 4, we can interpret that a one standard deviation contractionary monetary policy shock has a negative impact on real equity prices, almost immediately upon impact, however insignificant at the 68% level until 7 months. The findings are in line with Chatziantoniou, Duffy and Fillis (2013), who find that a contractionary monetary policy is associated with a lower stock return in the UK, with stronger effects when policy shocks are identified around decision windows. This emphasises the dominance of the discount rate channel in the short run; however, the lagged significant effect can be explained due to the cash flow channel that materialises with delays due to demand, margins and wages, as our identification is monthly. The cash flow component that makes the decline statistically significant shows only as firms update guidance and earnings.

The positive effect we see on Industrial production or output, which shows a 68% rise and persistence and a 5-month window of rise at the 95% level is a counterpart to the price puzzle, the policy innovations could be contaminated by the information the central bank has about stronger than expected demand, if the modelling does not purge this information by using daily instruments, the initial rise in output before the conventional contraction dominates. UK evidence shows that pure policy rate surprises eventually reduce activity and inflation, but the announcement windows contain information shocks that can move yield in the same direction (Cesa-Bianchi, Thwaites & Viconda, 2020).

In contrast to our results from the previous model, a contractionary monetary policy shock reduces inflation with lags; however, the effect is insignificant at the 68% confidence interval. Short-run CPI values can be affected by exchange rate pass-through under dominant currency pricing and staggered resets. The pass-through effect into UK consumer prices tends to be stronger when the exchange rate moves in response to monetary shocks (Forbes, Hjortsoe and Nenova, 2018) as the exchange rate appreciates, import prices ease, and headline CPI deflates.

For the exchange rates, we can see a sharp appreciation following a contractionary monetary policy shock, and it shows persistent effects over 30 months. SVAR evidence for advanced small open economies like the UK shows that unexpected rate hikes appreciate the currency on impact at longer horizons (Kearns & Manners, 2006). For the UK specifically, around monetary policy announcements, there is a sizeable exchange rate response to target and path surprises and with dominant currency pricing in trade, such appreciations transfer to import prices relatively swiftly, justifying the disinflation we see in the CPI

5.3 Recursive models Structural Error Variance Decomposition for Norway and UK

Period	S.E.	WIP	IPI	CPI	M3	POLICY	EXRATE	STOCKS
1	0.079362	0.000373	1.715274	6.176992	0.027817	0.580340	0.481686	91.01752
2	0.080980	1.214536	1.738249	6.535303	0.528500	1.422228	0.462660	88.09852
3	0.085200	4.750903	2.342817	9.087809	0.532827	3.100083	0.418049	79.76751
4	0.086221	4.651444	2.556564	8.873886	1.247882	3.454547	1.316829	77.89885
5	0.087355	4.562690	3.417906	8.704587	2.213881	3.367705	1.283138	76.45009
6	0.089484	5.191761	3.270905	8.295507	2.183664	5.927769	1.228993	73.90140
7	0.091156	5.020573	3.922330	8.663996	2.179846	7.409994	1.186673	71.61659
8	0.092097	5.327937	3.851299	8.509325	2.355831	8.391514	1.381608	70.18249
9	0.092449	5.290015	3.838847	8.458400	2.340202	8.981826	1.371499	69.71921
10	0.092806	5.323052	4.224477	8.410291	2.380865	9.107210	1.370890	69.18322
11	0.093008	5.396244	4.211195	8.563910	2.408352	9.169794	1.365066	68.88544
12	0.093095	5.464896	4.203333	8.562713	2.475497	9.167694	1.363236	68.76263
13	0.093149	5.469365	4.207623	8.553128	2.474657	9.158918	1.387314	68.74899
14	0.093205	5.464146	4.212293	8.547387	2.546231	9.150287	1.387408	68.69225
15	0.093257	5.486113	4.208154	8.590324	2.565182	9.140471	1.387747	68.62201
16	0.093303	5.484962	4.206635	8.643596	2.580705	9.137315	1.390725	68.55606
17	0.093325	5.483077	4.204732	8.679226	2.585176	9.133248	1.390153	68.52439
18	0.093338	5.485139	4.204422	8.683787	2.594107	9.135424	1.390059	68.50706
19	0.093343	5.486870	4.204694	8.683367	2.594476	9.137355	1.390604	68.50263
20	0.093349	5.486711	4.204876	8.685092	2.596239	9.140572	1.390445	68.49606
21	0.093353	5.486249	4.204533	8.685401	2.596481	9.147352	1.390465	68.48952
22	0.093357	5.486312	4.204642	8.685600	2.597554	9.150357	1.391511	68.48402
23	0.093359	5.486692	4.205102	8.685740	2.597579	9.152556	1.391457	68.48087
24	0.093361	5.487020	4.205232	8.686676	2.597514	9.153526	1.391526	68.47851

Table 3:SFEVD Norway

Turning to our SFEVD results from table 3: we can see a small contribution of 0.6% at the first horizon, 5.9% at horizon six before settling to 9.1% from horizon 10, as much of the immediate equity move is discount rate channel news that is shared with other shocks the impact is small at the early horizon and as the Cholesky identification bundles some of the same day effects into the exchange rate or global components. Here the standard errors represents the sampling uncertainty around the reported variance, for a specific shock and horizon. Here it reflects parameter estimation error.

At the longer horizons, policy's contribution accumulates as the expected short rate path dominates the discount rate for longer (Gurkayanak, Sack & Swanson, 2005). Equity shares explain most of their own variation at horizon one at 91%, under EMH as prices instantly absorb a wide information set, on the longer horizons once external demand and financial conditions explain a larger fraction of the equity fluctuations, therefore the own shock to equites falls to 68% around the 11 month horizon and settles as own variation and fluctuation remains significant.

Period	S.E.	WIP	IPI	CPI	M3	POLICY	EXRATE	STOCKS
1	0.054569	0.028517	0.000237	0.630047	0.005506	0.197785	0.102437	99.03547
2	0.055306	0.319508	0.457432	0.950114	1.156100	0.274017	0.307217	96.53561
3	0.057202	2.038819	3.288587	1.135581	1.112453	0.474284	0.861875	91.08840
4	0.057558	2.017942	3.408811	1.772994	1.236984	0.554385	0.934052	90.07483
5	0.057874	2.365870	3.572848	1.763169	1.296168	0.569563	1.333003	89.09938
6	0.059158	3.408658	3.443721	3.844735	1.243431	0.724053	2.030129	85.30527
7	0.059853	3.353155	3.484286	3.945371	1.897785	1.804365	2.130175	83.38486
8	0.060015	3.356857	3.758135	4.085590	1.889279	1.801146	2.175918	82.93307
9	0.060172	3.339585	3.739287	4.358393	1.881197	1.816999	2.172901	82.69164
10	0.060286	3.335678	3.769316	4.543642	1.892959	1.810194	2.187210	82.46100
11	0.060421	3.403217	3.753657	4.634757	2.003400	1.810377	2.186771	82.20782
12	0.060616	3.395904	3.814425	4.855451	2.032146	1.846490	2.179627	81.87596
13	0.060760	3.474155	3.838306	4.944750	2.023654	1.908187	2.174439	81.63651
14	0.060804	3.508238	3.861901	4.977964	2.021000	1.905459	2.174454	81.55098
15	0.060850	3.539121	3.867686	5.005631	2.018104	1.924210	2.188188	81.45706
16	0.060895	3.578508	3.883271	5.024255	2.016990	1.936967	2.190943	81.36907
17	0.060930	3.607787	3.892805	5.038767	2.016697	1.940170	2.193883	81.30989
18	0.060952	3.631642	3.894417	5.040521	2.019423	1.939164	2.198634	81.27620
19	0.060965	3.654032	3.899305	5.040227	2.021581	1.941117	2.197895	81.24584
20	0.060973	3.662593	3.902407	5.041557	2.023697	1.942475	2.198506	81.22877
21	0.060979	3.671214	3.902954	5.040702	2.027839	1.942282	2.199155	81.21585
22	0.060983	3.678374	3.904544	5.040383	2.030769	1.942224	2.198853	81.20485
23	0.060986	3.679827	3.904465	5.040073	2.033747	1.944551	2.198786	81.19855
24	0.060987	3.680867	3.904248	5.041250	2.036049	1.944906	2.198666	81.19401

Table 4: SFEVD UK

From table 4, the own shock of equities initially explains 99% of the forecast and stabilises at 81% at the 12-month horizon, similar to our previous observation. Policy also contributes to a small share of the predictive power of equities, starting at 0.2% and selling at 1.9% at the 13-month horizon. A possible explanation for this could be related to the index compositions and earning geography, the OBX index where the 25 most traded Oslo Bors stocks is industry concentrated in energy, finance, shipping and industrials, the top weights include Equinor (integrated oil & gas) and DNB (banks) with several other energy and commodity exposed firms whose cash flows and discount rates are tightly linked to the Krone, global demand and domestic credit conditions.

By contrast, the FTSE 100 is a globally tied index and considered to be less domestically sensitive than broader indices. Therefore, for the UK, the FEVD is smaller as more of its shocks have to compete with global and foreign drivers, as most of its volatility is explained by external shocks.

5.4 Model 3: Norway Baseline

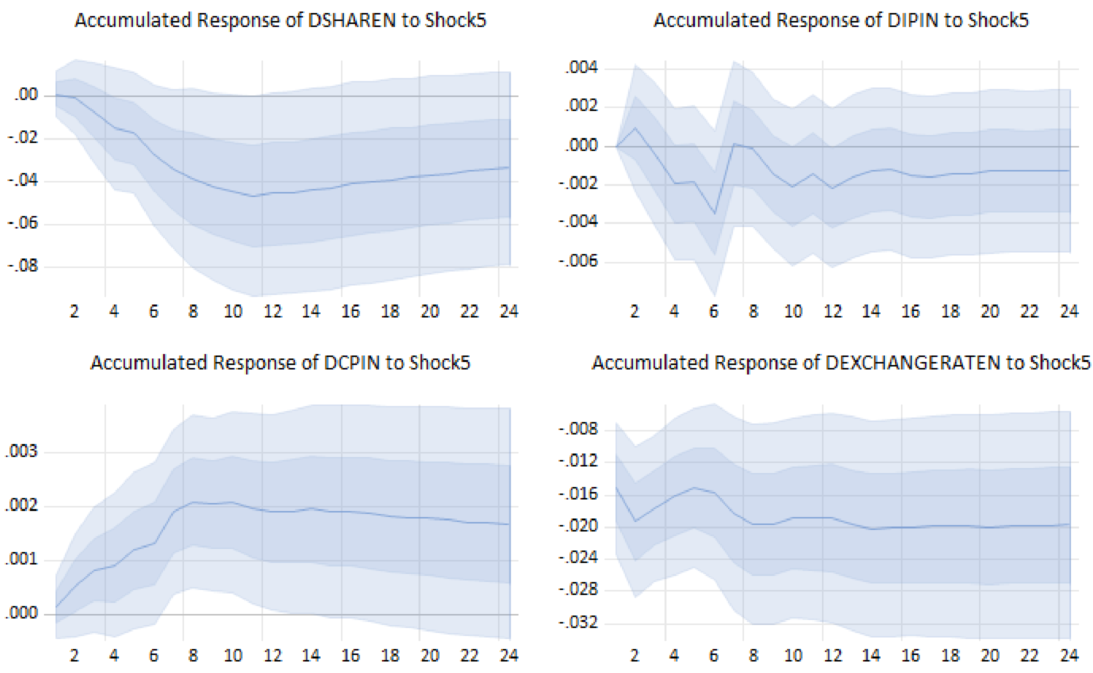

Figure 5: Impulse response function for Norway

From Figure 5, we can interpret that following a one standard deviation contractionary domestic policy shock still affects real equity prices negatively and is aligned with the literature; however, the response is smaller in magnitude and slower to reach statistical significance at approximately 4 months. This attenuation can be attributed to the fact that the information component of the policy announcement is separated from the pure target factor; the contemporaneous equity move need not be consistently negative if a part of the contractionary policy indicates weaker currency or lower inflation ahead, and that information can partly offset discount rate news in the announcement window (Jarocinski & Kardi, 2020).

Allowing monetary policy to move simultaneously with the exchange rate results in more of the immediate adjustment to the policy news being absorbed by the exchange rate, both through discount channels (global term premia) and cashflow channels (currency translation of foreign revenue), so less weight is placed on equities on impact (Bjornland, 2009). Cross-country evidence on Norway finds that the variance share attributable to domestic policy is modest once other macroeconomic factors have been accounted for.

The impact on industrial production remains similar to that of the recursive model; however, we observe a narrow, significant drop at the 6-month horizon, confirming our earlier discussion on the lagged effect on production. Placing the foreign rate first lets external monetary news relevant to Norway's import-oriented production and financial conditions absorb short-run movements that a domestic-only model could misattribute to Norwegian policy shock.

The impact on CPI is more pronounced in this model, while the recursive model suggested a mild and delayed disinflation, the non-recursive model displays a positive effect significant at the 95% confidence interval. In Norway's open economy setting, the NEER depreciates heavily on impact, displaying the overshooting pattern, but the sign is negative, raising import prices under dominant currency pricing and staggered pass-through. Empirically, this effect is shock-dependent and tends to be larger when exchange rates are driven by monetary or financial shocks.

The exchange rate response is where the contrast with the recursive ordering is most striking, especially in the case of Norway, where the exchange rate should appreciate following a contractionary monetary policy shock due to higher rates making domestic assets more attractive, capital inflow rises, and demand for domestic currency strengthens. However, as we see here, the exchange rate sharply depreciates.

In our model, when policy and exchange rate are allowed to interact contemporaneously, the exchange rate can exhibit overshooting driven by both relative expected short rates and time-varying risk premia, depending on the balance between a pure rate surprise and information on growth and risks, a depreciation could be possible. The use of a geometric trade-weighted NEER matters, as ECB and federal funds surprises co-move strongly The composition of the currencies could reveal a single bilateral rate that appreciates. Norway's placement as a commodity currency ties the Krone to oil-related cash flow news. When tightening episodes coincide with deteriorating oil prospects, the Krone tends to depreciate. The short-run restrictions permit these risk premiums and external news components to surface in the exchange shock rather than being passed through the policy equation.

However, similar identification, excluding foreign rates and including NEER allowing for contemporaneous movement with policy has been shown to have an appreciation (Inoue & Rossi, 2019; Bjornland, 2009), therefore, we cannot completely reject this effect as an inaccurate artefact of the model and the most plausible explanation remains to be the inclusion of foreign rate, as we do not observe this in the recursive model.

5.5 Model 4: UK Baseline

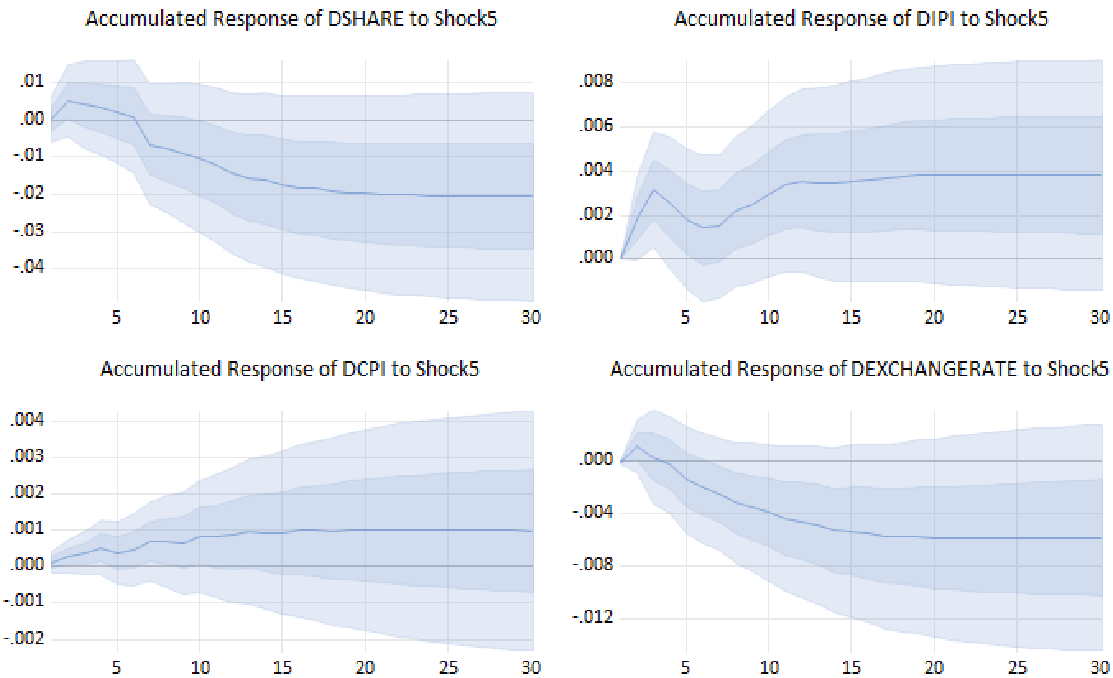

Figure 6: Impulse response function for the UK

From Figure 6, we can infer that the UK equity market holds the expected negative impact from a one standard deviation contractionary monetary policy shock; however, unlike the recursive model, the real equity prices are significantly negative after 7 months and maintain a persistent effect thereafter; the 95% level is wider and overlaps zero for the duration of the shock. The delayed significant effect can be attributed to the information shocks in announcements, as in the case of our Norway Baseline model.

The cash flow offset in an open economy, resulting from currency depreciation as found in our model, can increase foreign currency revenues in GBP for internationally reliant FTSE constituents, making the initial fall in real equity prices statistically insignificant. As tighter monetary conditions filter through demand, credit, and wage costs, the cash flow channel turns negative, and the decline becomes statistically significant.

The activity puzzle can also be attributed to the information effect, as news on demand is stronger than expected, the identified policy shock inherits parts of the favourable demand news unless high-frequency instruments fully remove it. The immediate outcome is the stronger outcome in the subsequent years, despite the shock being contractionary when isolated. This effect is identified by Nakamura & Steinsson (2018) in their study; however, the effect is much delayed than we find in our study, therefore, it cannot be completely attributed to pure information effect shocks. We observe a smaller, yet significant, persistence compared to the recursive model, where output was not restricted from affecting monetary policy contemporaneously, which strengthens the choice of restrictions used.

For CPI we see a median rise and gradual fall over a year, however the effect is only negligently significant at the 68 % level for the initial months following a contractionary monetary policy shock, this initial and weakly significant effect can be contributed to the exchange rate depreciation as identified shock includes a sizeable information or risk premium component that weakens the Sterling and supports external demand the disinflation from weaker domestic demand appears gradually with lags, leaving the early positive CPI response.

Turning to the exchange rate, we see a depreciation following a contractionary monetary policy shock. When contemporaneous policy and foreign exchange feedback is permitted, the exchange rate equation can load not only on the relative short rate path but also on time-varying exchange rate premium, global term premium, news and information shocks, which can move the exchange rate independently of the policy target surprise. Evidence from Li, Iscan & Xu (2010) shows that in the case of Canada, there is an expected appreciation in the short run, but depreciation in the medium-run dynamics.

5.6 Baseline Structural Forecast Error Variance Decomposition of Norway and the UK

Period	S.E.	FR	IPI	CPI	M3	POLICY	EXRATE	STOCKS
1	0.082157	0.001880	1.665721	6.368244	0.202539	0.016696	0.472646	91.27227
2	0.083651	0.341365	1.728661	6.832545	0.875319	0.056073	0.467035	89.69900
3	0.086198	1.683584	2.150719	8.860027	0.948345	0.699902	1.113412	84.54401
4	0.087501	1.956783	2.478215	8.598561	2.144688	1.403877	1.297119	82.12076
5	0.088427	1.919087	3.215066	8.501360	3.084058	1.427368	1.307864	80.54520
6	0.089920	1.871508	3.118571	8.242540	3.031368	2.714180	2.338308	78.68353
7	0.091390	1.813847	3.690345	8.870926	3.322688	3.154657	2.790659	76.35688
8	0.092028	2.112604	3.684806	8.751008	3.540209	3.326420	3.277177	75.30778
9	0.092368	2.454517	3.658792	8.759700	3.514244	3.476445	3.378395	74.75791
10	0.092573	2.485620	3.898805	8.758577	3.545483	3.519581	3.364630	74.42730
11	0.092783	2.819220	3.883730	8.766978	3.533782	3.552615	3.349420	74.09425
12	0.092853	2.894120	3.879557	8.754153	3.531405	3.561383	3.365627	74.01376
13	0.092971	3.034814	3.881798	8.734900	3.525884	3.554783	3.404304	73.86352
14	0.093025	3.095138	3.885770	8.727189	3.529787	3.567709	3.415910	73.77850
15	0.093086	3.114582	3.881902	8.744752	3.526612	3.575916	3.462488	73.69375
16	0.093162	3.172189	3.876339	8.748544	3.520920	3.610668	3.478597	73.59274
17	0.093196	3.179399	3.876001	8.766533	3.519185	3.613480	3.495701	73.54970
18	0.093226	3.189133	3.873613	8.768723	3.517691	3.631164	3.511069	73.50861
19	0.093241	3.196626	3.873136	8.766234	3.517214	3.638206	3.520752	73.48783
20	0.093257	3.195858	3.872328	8.763367	3.516223	3.652988	3.534607	73.46463
21	0.093269	3.197291	3.871307	8.761070	3.515296	3.660924	3.545996	73.44812
22	0.093279	3.197319	3.870500	8.759209	3.514742	3.671880	3.553937	73.43241
23	0.093287	3.196804	3.870000	8.758085	3.514156	3.677736	3.562498	73.42072
24	0.093292	3.196963	3.869719	8.757321	3.514098	3.682205	3.566155	73.41354

Table 5: SFEVD Norway

From the error variance in table 3, we can see a small contribution and explanation at the first horizon of 0.01% and stabilising at the horizon 16 at 3.6% this contrasting reduction in explanatory power is contributed to by the simultaneous impact of foreign rate and exchange rates. Interestingly, the contribution to the explanation of stocks remaining relatively unchanged and the CPI explains the largest variance in real equity prices in the model. In a small open economy, a shock-dependent pass-through can cause the headline CPI to rise temporarily after monetary shocks, thereby deflating stocks and increasing the contribution of CPI's explanatory power at short horizons, rather than allocating it to policy.

Period	S.E.	FR	IPI	CPI	M3	POLICY	EXRATE	STOCKS
1	0.055265	1.337290	0.015378	0.104746	0.355287	0.001013	0.402894	97.78339
2	0.056124	2.198019	0.479053	0.167489	0.345608	0.793953	1.204219	94.81166
3	0.057073	2.151166	1.264061	1.188704	0.343277	0.804197	1.369675	92.87892
4	0.057470	2.541104	1.309481	1.226793	1.014075	0.820273	1.454264	91.63401
5	0.058543	5.096882	1.845737	1.380858	1.012857	0.826356	1.524114	88.31320
6	0.059393	4.961064	1.869369	1.995316	2.639026	0.839710	1.795120	85.90040
7	0.060163	5.123784	1.868592	2.249737	2.679208	2.359782	1.974455	83.74444
8	0.060215	5.129858	1.866284	2.249183	2.719656	2.384165	1.980658	83.67019
9	0.060416	5.215129	1.863640	2.279801	2.907628	2.416290	2.111564	83.20595
10	0.060492	5.202895	1.873818	2.285201	2.977939	2.473049	2.108079	83.07902
11	0.060581	5.242507	1.886757	2.280850	2.992682	2.526098	2.109264	82.96184
12	0.060670	5.234746	1.889768	2.292218	3.011147	2.674319	2.107977	82.78982
13	0.060749	5.286211	1.887244	2.286447	3.038487	2.715629	2.133711	82.65227
14	0.060768	5.284212	1.886950	2.293679	3.038546	2.717179	2.137201	82.64223
15	0.060787	5.289930	1.886371	2.292321	3.041003	2.754934	2.135980	82.59946
16	0.060799	5.292977	1.885629	2.292583	3.039987	2.777024	2.135180	82.57662
17	0.060809	5.292727	1.885962	2.296386	3.042026	2.777870	2.135969	82.56906
18	0.060817	5.292026	1.886296	2.296926	3.041478	2.785916	2.135880	82.56148
19	0.060820	5.294999	1.886497	2.297043	3.041169	2.790808	2.135695	82.55379
20	0.060822	5.295098	1.886437	2.298733	3.041039	2.791297	2.136635	82.55076
21	0.060824	5.296028	1.886367	2.298922	3.041009	2.793730	2.136525	82.54742
22	0.060824	5.296167	1.886332	2.299102	3.041914	2.795005	2.136499	82.54498
23	0.060825	5.296176	1.886417	2.299779	3.041875	2.795008	2.136565	82.54418
24	0.060825	5.296204	1.886425	2.299918	3.042365	2.795339	2.136536	82.54321

Table 6: SFEVD UK

From Table 4, we see that equity's own explanatory power dominates in the short and long run and remains stable at 82.5%. We also see monetary policy’s explanatory power remains low and stabilises around 6 month horizon, large contributions from exchange rate and prices still persist at longer horizons and contribute a larger explanatory power together compared to monetary policy, consistent with our previous assumption on why monetary policy explanatory power in Norway is marginally larger than in the UK due to the globally exposed nature of the sectors.

Our baseline findings align with Li iscan and Xu, 2010 where they find the explanatory power policy to stocks in Canada is 5.2% and the US 3.1% is which are similar to our findings and provide evidence when comparing countries and using the same methodology we get a similar conclusion of a very small variance of the volatility and movement in real equity prices are explained by monetary policy.

5.7 Foreign monetary policy on stocks in the UK and Norway

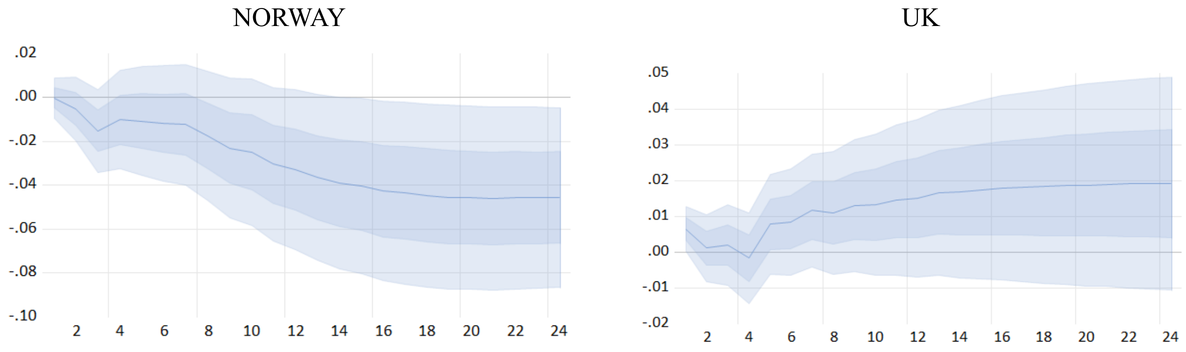

Figure 7: Impulse response functions for Norway and the UK

From Figure 7, we find an interesting result on the impact of foreign policy on domestic real equity prices. Figure 5 shows the effect of a one standard deviation contractionary foreign monetary policy on real equity prices in Norway that turns immediately negative and is significant on the longer horizons at the 95% level. A US monetary policy contraction raises global discount rates and tightens global financial conditions, following the global financial cycle, lowering foreign risky asset value (Agrippino & Rey, 2020).

The commodity and currency link provides an explanation for this effect; a contractionary federal funds rate and a stronger purchasing power of the dollar coincide with lower oil prices through weaker global demand and tighter financial conditions. For commodity-dependent currencies like the Norwegian Krone, oil prices can be predictive for the exchange rate and cash flows in the oil industry (Ferraro, Rogoff and Rossi, 2015); therefore, lower prices and higher global discount rates depress the present value of Norwegian energy sector earnings and the broader index. Further supported by our cited literature on the impact of oil prices on Norwegian equity shares (Aastveit, Bjornland and Thorsrud, 2015). In agreement with the higher risk premia as the US rate rises, it increases the price of risk and compresses intermediary balance sheets internationally, which increases required returns and lowers equity valuations abroad.

The difference, as we can see from Figure 6, for the UK, following a one standard deviation contractionary foreign policy rate, the real equity prices fall slightly in the first 4 months, then rise persistently in the months after, at the 68% level, insignificant at the 95% level. Although this seems like an inverted sign convention, Li, Iscan & Xu (2010) find the same effects for

Canada, although this effect is not persistent in the long run. Their finding is a result of the Canadian stock prices being strongly financially integrated with the USA therefore, the UK results are not directly comparable to it, however the justification provided holds for the UK as they emphasise that trade channel shocks are relatively muted for stock prices as nominal exchange rate movements cushion the adverse shifts in foreign demand, therefore exchange rate adjustments help offset some of the weight placed on equites in the impulse responses, this results in a short lived positive equity response to a US rate hike that fades as broader contractionary indirect effects channel through.

For the FTSE's globally oriented constituents, and consistent with the explanatory results from our study, a sterling depreciation lifts the GBP-reported earnings, reversing the discount rate effect on stock valuations. As noted, US policy shocks transmit quickly to foreign asset prices (Dedola, Rivolta & Stracca, 2017), while UK-focused studies show shock-dependent pass-through from exchange rate to pass-through from domestic prices, consistent with revenue translation effects (Forbes, Hjortsoe and Nenova, 2018).

In conjecture with the global financial cycle, US policy shifts can reorganise global risk premia and capital flows. Depending on the risk adversity, conditions can favour UK large caps with international earnings, which is why we only see a small magnitude of the effect and wide confidence intervals, suggesting an eventual long-run fall and in fact, we see this effect when testing alternative lag specifications and modelling.

Chapter 6: Discussion

6.1 Interpretation of results

In a rational asset-pricing framework, the real equity price equals the expected present value of future real cash flows discounted by the stochastic discount factor. A monetary tightening that lifts the expected path of real short rates and can also increase required risk premia, raise the discount factor and lower present values on impact, before firms have time to revise dividends or investment Using real equity returns removes nominal price-level noise; therefore, at monthly horizons and under recursive identification, the immediate equity response is naturally interpreted as discount-factor news rather than concurrent cash-flow changes. This could explain the immediate sharp drops in the median value that we see in the recursive model, as the tightening is mapped directly onto the discount factor.

A complementary theoretical standpoint from the results is the risk-taking channel; contractionary policy reduces financial intermediaries' balance sheet capacity and raises the market price of risk. In a present value identity, this appears as a higher equity risk premium with the additional higher expected path of real short rates, both having a negative impact on equities. From our results, the risk-taking channel is plausibly stronger in Norway as its financial system is more bank-dominated (Onega, Smith and Michalsen, 2003). The explanatory power of monetary policy is therefore higher as rate increases affect credit supply more directly, raising risk premiums. The UK's listed market is more internationally diversified (Hardie et al., 2013) in both currency and sector composition, which affects the contemporaneous premium rises in real equity valuation and reduces statistical significance at the earlier months.

In accordance with the findings of Li, Iscan, and Xu (2010), monetary policy provides a small contribution in terms of explanatory power of stock variation in the short and long run, under informational efficiency, where the news that was not present in the previous period, such as earnings surprises, guidance changes, mergers, etc., a small macro-economic SVAR cannot capture these informational effects successfully and contributes to the stock's own structural disturbance. Additionally, in a New-Keynesian general-equilibrium setting with a Taylor-type rule, a contractionary policy shock raises the nominal policy rate and, through credibility and expectations, tends to lower expected inflation, resulting in a rise in the real rate Because the policy rule is mean-reverting around the target (and inflation expectations are anchored), the expected deviation of real short rates is often limited in persistence.

Hence, the present-value effect on equities is correspondingly modest when rate revisions are transitory; large equity moves require persistent revisions to the expected path of real rates or to risk premia. This framework helps rationalise why, in our recursive monthly SVAR, the equity response drops immediately on impact, suggesting a discount-factor news effect, as mentioned, while the overall explanatory power of monetary policy for equity variation remains limited.

On the cash flow side, sticky prices, markup smoothing, and global supply chain competition lead to slow and muted dividend responses to a one-off policy shock; most dividend variation is driven by technology, sectoral shocks, and global demand rather than domestic policy General equilibrium offsets further shrink the net effect; therefore, higher discount rates push prices down, but the same announcement can convey good news information about growth or inflation control, partly raising expected cash flows. In open economies, the exchange-rate channel and foreign financial conditions absorb part of the shock, diluting domestic impact. Finally, under rational expectations, most of the ruling component of policy is anticipated and already priced; only the surprise can move equities.

The effect of foreign contractionary monetary policy affects domestic equities initially through the global discount rate channel. In present value terms, the higher policy rates lift the global risk-free path and, via constrained intermediaries, raise required risk premia; both of which increase the discount factor and lower valuations abroad before local cash flows move (Gabaix & Maggiori, 2015).

In open economy New-Keynesian models with incomplete international risk sharing, foreign monetary shocks shift the stochastic discount factor through pricing-kernel co-movements; therefore, domestic asset prices respond under floating rates (Devereux & Sutherland, 2010). This is suggestive of the fact that the effect we see in the Norway baseline model is more aligned with the theory and hence shows a significant negative impact in the long run at the 95% confidence interval, while the UK model remains insignificant at the 95% confidence interval. This delayed effect that our model does not capture is further supported by portfolio balancing and exchange rate asset pricing. With domestic bias and infrequent portfolio adjustment, a foreign tightening influences wealth and rebalancing flows that make equity prices and exchange rates co-move. Outflows lower equities and weaken the currency, while safe assets (negative conditional beta to global risk) can have the opposite effect (Hau & Rey, 2006).

In asset market general equilibrium models, the same shocks that move exchange rates affect the equity prices through demand and terms of trade (Pavlova & Rigobon, 2007). Therefore, a depreciation can raise domestic currency revenues for globally earning firms, partly offsetting the rise in the discount rate; conversely, if the foreign contraction coincides with adverse news on terms of trade, a depreciation does not cushion equities. In the baseline specification, policy and exchange rate feedback allow some of the same window repricing to be absorbed by the exchange rate (risk premium and portfolio balance news), and the equity response is delayed.

The results imply systematic cross-country heterogeneity. International Capital Asset Pricing theory suggests that currency risk is priced in equities; therefore, markets with high foreign revenue shares or natural currency hedges can benefit from depreciation in reported earnings. While commodity exporters experience stronger global conditions, they also have lower commodity-linked cash flow (Adler & Dumas, 1983). Due to infrequent portfolio decisions, exchange rate and equity co-movements can be persistent, signifying cross-country differences (Bacchetta & Wincoop, 2010), and much of the exchange rate move reflects time-varying risk premia rather than uncovered interest parity (Engel, 2016). This is why the UK equities, which are globally diversified, show a statistically muted or positive initial response, while Norway’s energy and bank-reliant market displays a larger and significant negative effect when global premia and terms of trade news dominate.

6.2 Robustness and alternative models

We estimate the Kilian bootstrap inference on the reduced form VAR. It resamples the VAR residuals and computes responses under a fixed identification based on the Cholesky ordering. This provides percentile confidence bands robust to non-Gaussian errors and small samples. This supports the recursive model findings, an immediate negative equity mean response and delayed quantity and price effects (Appendix, Figure A3, A4)

As the literature suggests, Norway's stock market can be heavily influenced by oil trade fluctuations therefore, we estimate the recursive model with oil as an exogenous control replacing world industrial production, for both the UK and Norway (Appendix, Figure A5, A4; Table A13, A14). The IRFs remain unchanged in the context of the period of effect and sign, suggesting our choice of global economic control sufficiently captures the fluctuations. The share of explanatory power also remains similar. For Norway, the stocks' own shares rise from 68.8% (WIP) to 70.4% (Brent), and the policy share falls from 9.17% to 8.72% at the 24-month horizon; other contributions move within half a percentage point. In the case of the UK, the IRFs also maintain timing and sign conventions for explanatory power. Equity shares explain 79.3% (Brent) and 81.1% (WIP), while policy accounts for 1.9% (WIP) to 1.6% (Brent).

For our baseline model, we include WIP in the model; the IRFs for real equity responses remain unchanged (Figure A7, A8; Table A15, A16) for the UK and Norway. However, in terms of explanatory power, the policy goes from 2.7% to 3.1% (including WIP) in the UK model and from 3.6% to 5.2% (including WIP) in the Norway model, suggesting more sensitivity to global factors. We choose to report the baseline without WIP due to the small changes and to maintain model stability, as including WIP can prevent it from converging at conventional iterations.

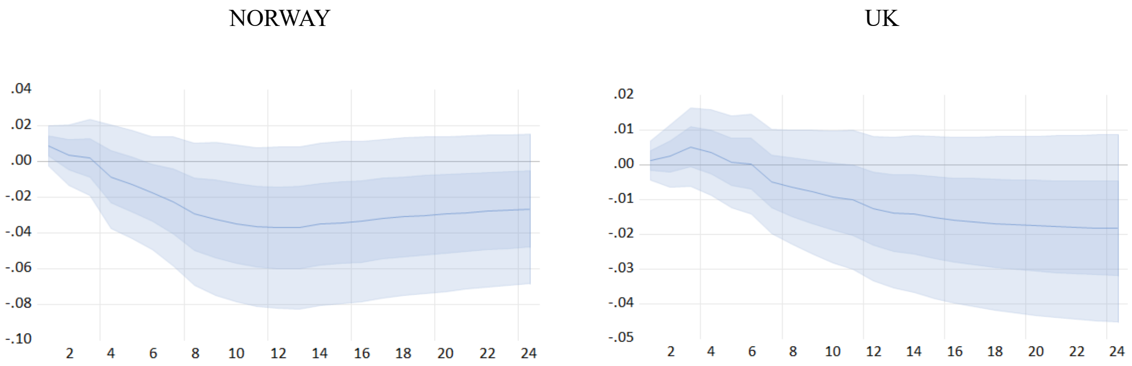

Figure 8: Impulse response function for Norway and the UK

We estimate the baseline model using the OECD share price index to mitigate concerns about sector composition heterogeneity ensuring the sign conventions remain unchanged. As a result, we observe that the significance of the negative effect is delayed; however, both models yield similar outcomes to our baseline. While it does not address sector-level heterogeneity, it supports our claim that contractionary monetary policy has a generally negative impact on stocks.

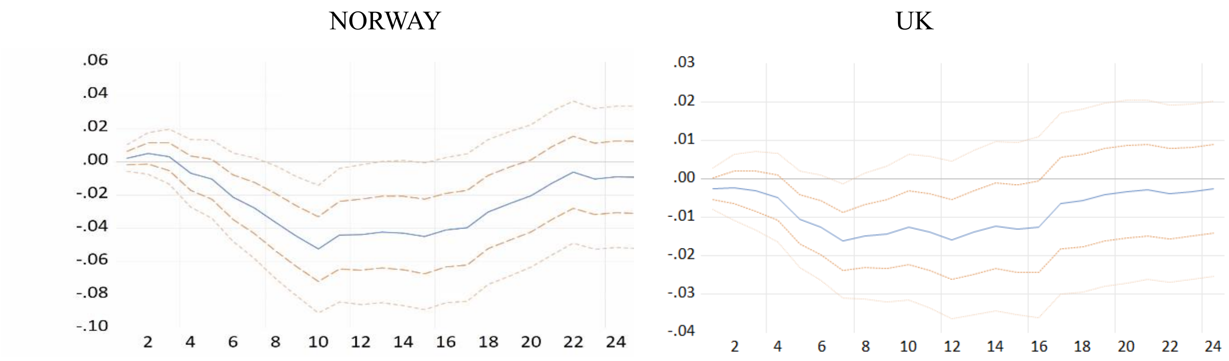

Figure 9: Impulse response function for Norway and the UK at 12 Lags, monetary policy impact on stocks

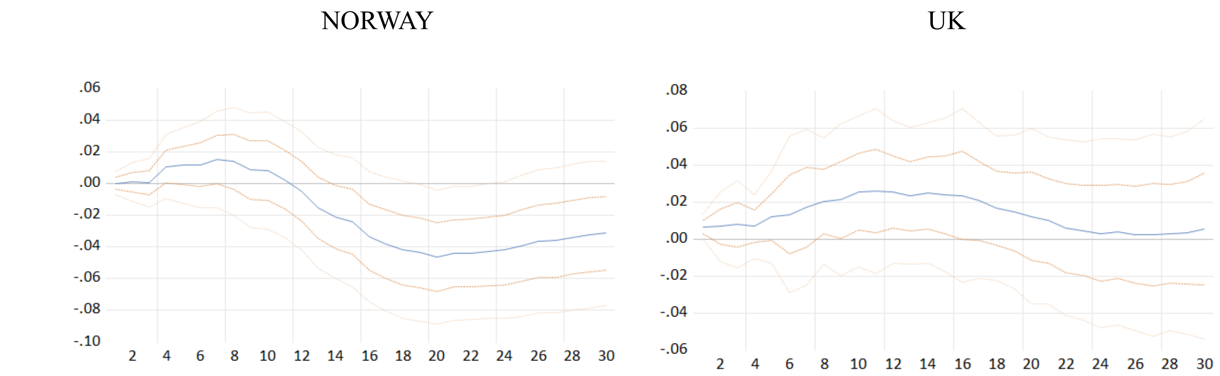

Figure 10: Impulse response function for Norway and the UK at 12 Lags, foreign policy impact on stocks

In our 12-lag estimation of the model, where the model is covariance stationary and stable, and shocks represent a convergent moving average representation. The longer lag length captures annual seasonality. From a contractionary monetary policy shock, we see a clear mean reversion; the initial discount rate-driven decline fades eventually, and the cumulative effect returns towards zero over the medium run. There is no statistically significant permanent level change in equities, as under rational expectations, policy does not alter permanent real cash flow growth; therefore, long-run neutrality of money is sufficiently captured in our baseline specification. The negative effect is also significant for a small time period in both countries at the 95% confidence interval, supporting our claim of a negative impact on equities statistically.

For the foreign policy shock in our baseline report, the effect on the UK we find a significant permanent effect at the 68% level. The key adjustment margins, such as EX passthrough to import prices, earnings translation for globally listed firms, appear with lags. With 12 lags, those slower channels are explicitly accommodated. The UK equity response to a foreign tightening subsequently bends back toward zero, aligning with Li, İşcan and Xu (2010), who document that foreign shocks can look small or even favourable on impact but do not deliver persistent equity gains. A foreign contractionary policy that lifts global risk-free rates and risk premia should result in long-run negative equity responses, which our baseline model does not capture.

NORWAY UK

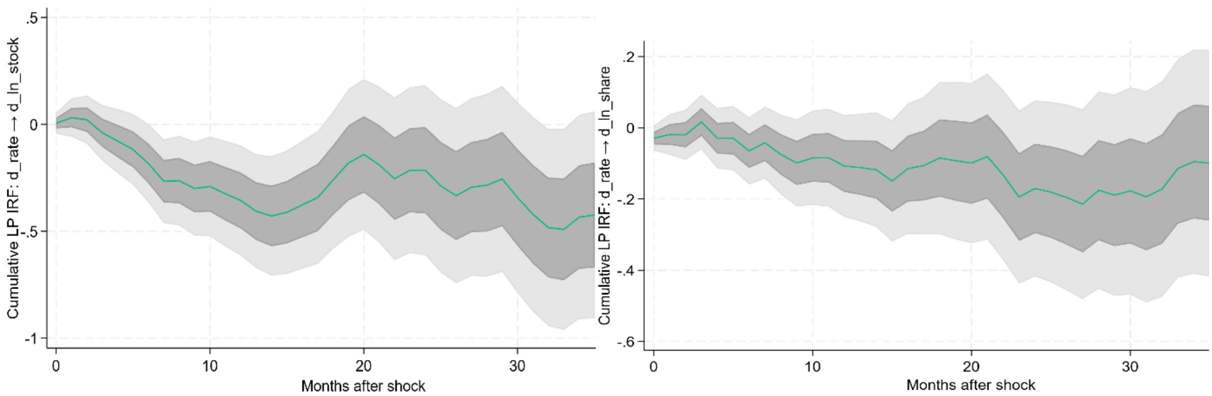

Figure 11: Localised plots for Norway and the UK, monetary policy impact on stocks

NORWAY UK

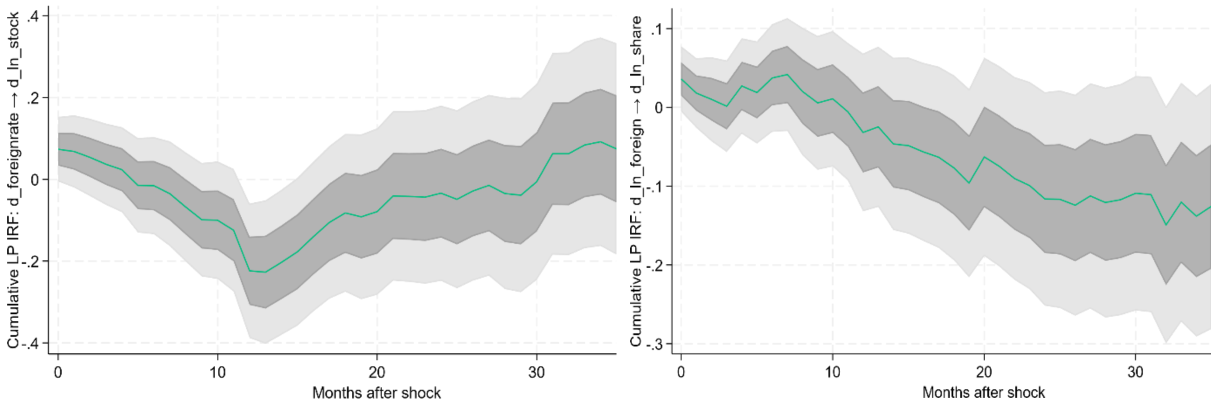

Figure 12: Localised plots for Norway and the UK, foreign policy impact on stocks

The expected negative impact on equities of contractionary domestic monetary policy is also captured, but the localised plots further validate our findings. Interestingly, horizon-specific regressions uncover the expected negative impact on equity prices in the UK in the long run. Confirming our earlier discussion on the present value framework, a foreign policy contraction raises the expected global rate and required risk premia, thereby increasing the real stochastic discount factor and reducing domestic equity valuations prior to cash flow adjustment. As the LP plots indicate, the response is faster in Norway bank-centred intermediation, widespread ARMs, and oil and krone exposure transmit foreign rate shocks quickly to discount rates and cash flows, while in the UK the FTSE's global revenue base and sterling-translation effects initially cushion valuations, pushing the negative adjustment to later horizons as confirmed by heterogeneous effects of foreign policy spillovers.

6.3 Limitations

A number of limitations of the study should be addressed before the results are considered for addressing policy implications. The exchange rate measure (NEER) may suppress policy-relevant signals as a trade-weighted exchange rate mix, bilateral policy surprises with commodity and time-varying risk-premium news; therefore, it can yield a depreciation, making it an exchange rate puzzle. A better alternative could be using USD or Euro-specific bilateral rates when comparing regions to consider a common currency and a risk premium proxy.

Several core IRFs are significant at the 68% but not at 95% This is common in a monthly SVAR small-sample bootstrap inference with non-Gaussian residuals, which produce fat-tailed IRF distributions and wider 95% bands; overlapping horizons also inflate variance (Kilian & Lütkepohl, 2017). High-frequency identification could be used to purge central-bank information effects, so monthly policy shocks do not blend target and path surprises with outlook news (Jarociński and Karadi, 2020).

The model specification may suppress long-run dynamics. Estimating differences in risks, looking at risks of losing cointegration relations among macro-economic variables, accumulated responses only partially recover this information. A levels VAR with breaks normalising IRFs to fixed basis point shocks for comparability and measuring ZLB/QE stance with a shadow short rate would strengthen interpretation. (Kilian and Lütkepohl, 2017).

Using only the U.S. policy rate omits the EUR policy, term-premium, QE shocks, risking contamination of domestic shocks, and headline indices embed currency-translation and sectoral exposures that can mask policy effects. A richer foreign block or a small GVAR treating the Euro area as exogenous and firm-level or revenue-weighted portfolios would better absorb external discount-rate and risk-premium news and isolate domestic transmission. (Miranda-Agrippino & Rey, 2020).

6.4 Policy implications

Our SFEVD results indicate that monetary policy explains a small share of stock market variation in the medium to long run and an even smaller variation in the short run, consistent with evidence for the US and Canada (Li, Iscan and Xu, 2010) and for the UK, where equity responses are short and time varying (Chatziantoniu, Duffy andFillis, 2013). As most equity movements reflect changes in the discount rate, tools such as counter-cyclical capital buffers and sectoral risk-weight analysis are preferable to rate changes when addressing asset price movements, with monetary policy focused on inflation and economic activity (Borio & Zhu, 2012). Additionally, when we implement theoretically plausible short-run restrictions, we find that the significant effect only appears in the medium to long-run dynamics, further supporting this suggestion.

External spillovers and surprise tightening are transmitted contemporaneously to domestic equities and exchange rates (Dedola, Rivolta, and Stracca, 2017). Policy projections and communications should incorporate the expected path of foreign rates and highlight the degree of uncertainty around global discount rate spillovers.

Asset prices react strongly to path (guidance) news while announcement winds often bundle outlook information with the policy surprise. Statements and projections should clearly distinguish between policy shocks and macroeconomic information, thereby reducing the misattributed influence of exchange rates or prices.

Given the weak short-run equity relation with policy, implementation should emphasise channels that move discount factors and cash flows reliably, for example, deposits and the credit channel via bank funding (Drechsler, Savov and Schnabl, 2017). Contractionary policy may not guarantee appreciation in the short run when risk premia or terms of trade shift (Li etal).

As equities capture only a fraction of the monetary policy transmission channel, credit spreads, term premiums, and lending standards could be valuable indicators, and modelling should include foreign policy and commodity trade shocks, especially when there is evidence of export revenues and exchange rate co-movement (Ferarro, Rogoff & Rossi, 2015).

Chapter 7: Conclusion

This study evaluates the response of the UK and Norwegian equity shares to domestic and foreign monetary policy using monthly data from January 1997 to December 2023. The findings of this study are in line with the literature, showing a negative impact of a contractionary domestic monetary policy on stock prices, in both our model specification and find no persistent effect when using longer lags. The sign is verified across specifications and remains consistently negative.

We also find a heterogeneous effect of foreign spillovers in our results, suggesting the need to include or control for global shocks and policy when analysing the domestic impacts of monetary policy. The explanatory power of domestic monetary policy on stock variation is found to be minimal across specifications, especially in the short run, which agrees with the overall consensus that monetary policy plays a limited role in determining stock prices. Policy makers should not rely on monetary policy as a primary tool for steering equity markets; rather, financial stability policies are better suited. In contrast, domestic decisions should explicitly account for time-varying foreign shocks and the exchange rate channel.

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Appendix

Table A1: Correlation of transformed variables

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) d_ln_ipi_uk	1.000							
(2) d_ln_cpi_uk	-0.008	1.000						
(3) d_ln_m3_uk	-0.119	-0.126	1.000					
(4) d_rate_uk	0.130	0.116	-0.162	1.000				
(5) d_ln_ex_uk	0.129	0.031	-0.149	0.302	1.000			
(6) d_ln_ftse_uk	0.027	-0.027	-0.131	0.019	-0.004	1.000		
(7) d_ln_oecd_uk	0.090	-0.026	-0.077	0.117	-0.029	0.682	1.000	
(8) d_foreignrate	0.108	0.175	-0.193	0.429	0.153	0.157	0.244	1.000

Table A2: Correlation matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) d_ln_ipi_no	1.000							
(2) d_ln_cpi_no	0.099	1.000						
(3) d_ln_m3_no	-0.085	0.050	1.000					
(4) d_rate_no	0.014	0.068	-0.060	1.000				
(5) d_ln_ex_no	0.048	0.020	-0.072	0.086	1.000			
(6) d_ln_obx_no	-0.080	-0.171	0.090	-0.138	0.165	1.000		
(7) d_ln_oecd_no	-0.065	-0.118	-0.056	-0.152	0.339	0.647	1.000	
(8) d_foreignrate	-0.048	0.065	0.069	0.137	0.102	0.219	0.274	1.000

Table A3: The results of the Augmented Dickey Fuller Test for the variables of the UK					
Variable	Test Statistics	1% Critical	5% Critical	10% Critical	Result
Transformed Output	-17.774***	-3.454	-2.877	-2.57	Stationary
Transformed CPI	-14.628***	-3.455	-2.878	-2.57	Stationary
Transformed Money Supply	-15.060***	-3.454	-2.877	-2.57	Stationary
Transformed Share Prices	-17.123***	-3.455	-2.878	-2.57	Stationary
Transformed WIP	-14.438***	-3.455	-2.878	-2.57	Stationary
Transformed Policy Rates	-9.923***	-3.454	-2.877	-2.57	Stationary
*Note: *** p<0.01, ** p<0.05, * p<0.1*					

Table A4: the results of the Augmented Dickey Fuller Test for the variables of Norway					
Variable	Test Statistics	1% Critical	5% Critical	10% Critical	Result
Transformed Output	-24.752***	-3.454	-2.877	-2.57	Stationary
Transformed CPI	-14.685***	-3.455	-2.878	-2.57	Stationary
Transformed Money Supply	-23.257***	-3.454	-2.877	-2.57	Stationary
Transformed Share Prices	-14.421***	-3.455	-2.878	-2.57	Stationary
Transformed WIP	-14.438***	-3.455	-2.878	-2.57	Stationary
Transformed Policy Rates	-8.404***	-3.454	-2.877	-2.57	Stationary
*Note: *** p<0.01, ** p<0.05, * p<0.1*					

Figure: A1 Roots inside the matrix for the UK at Lag 6

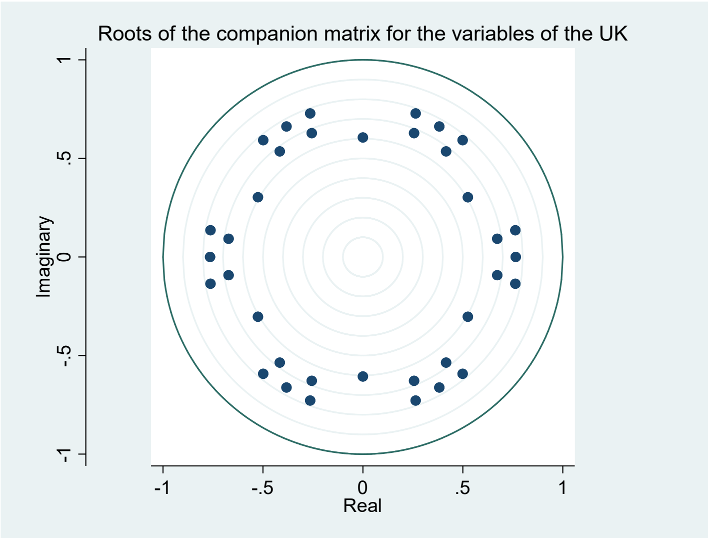

Figure: A2 Roots inside the matrix for Norway at Lag 6

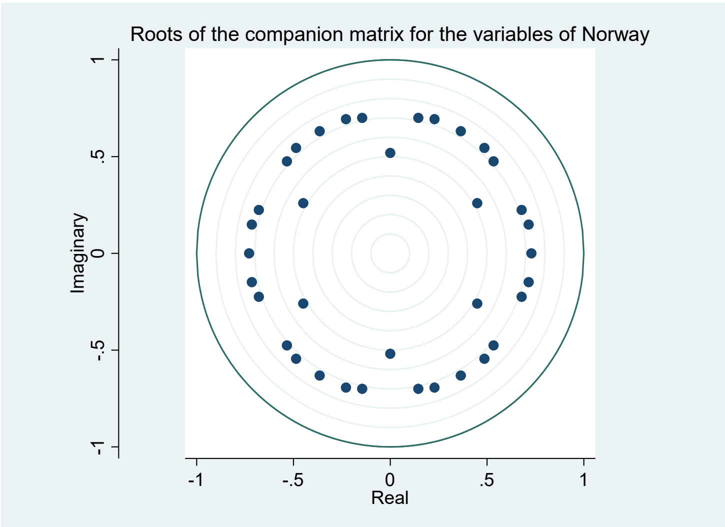

Table A5: Lag selection criteria for Norway

Lag	LogL	LR	FPE	AIC	SC	HQ
0	4781.813	NA	5.95e-23	-31.31025	-31.22487	-31.27610
1	4946.309	320.3630	2.79e-23	-32.06760	-31.38453*	-31.79439*
2	5014.547	129.7631	2.46e-23*	-32.19375*	-30.91299	-31.68147
3	5046.102	58.55900	2.76e-23	-32.07936	-30.20091	-31.32802
4	5079.397	60.25721	3.07e-23	-31.97637	-29.50023	-30.98597
5	5117.259	66.78618*	3.32e-23	-31.90334	-28.82950	-30.67387
6	5143.877	45.73179	3.86e-23	-31.75657	-28.08505	-30.28804

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Table A6: Lag selection criteria for the UK

Lag	LogL	LR	FPE	AIC	SC	HQ
0	5509.493	NA	5.04e-25	-36.08192	-35.99654*	-36.04777
1	5617.858	211.0450	3.42e-25	-36.47120	-35.78813	-36.19799*
2	5683.017	123.9095	3.07e-25	-36.57716	-35.29640	-36.06488
3	5746.193	117.2368	2.80e-25*	-36.67012*	-34.79166	-35.91877
4	5783.716	67.91191	3.03e-25	-36.59486	-34.11872	-35.60446
5	5817.868	60.24124	3.35e-25	-36.49750	-33.42366	-35.26803
6	5860.813	73.77988*	3.51e-25	-36.45779	-32.78627	-34.98926

* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

Table A7: Autocorrelation test for Norway

Null hypothesis: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	66.76906	49	0.0464	1.375424	(49, 1024.9)	0.0466
2	58.32201	49	0.1700	1.196539	(49, 1024.9)	0.1702
3	62.26583	49	0.0966	1.279879	(49, 1024.9)	0.0968
4	50.53173	49	0.4128	1.032835	(49, 1024.9)	0.4131
5	63.63190	49	0.0781	1.308819	(49, 1024.9)	0.0782
6	52.34304	49	0.3456	1.070789	(49, 1024.9)	0.3459

Table A8: Autocorrelation for UK

Null hypothesis: No serial correlation at lag h						
Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	66.94003	49	0.0451	1.380140	(49, 943.6)	0.0452
2	44.02393	49	0.6746	0.896874	(49, 943.6)	0.6750
3	63.87858	49	0.0751	1.314914	(49, 943.6)	0.0753
4	36.31288	49	0.9105	0.736818	(49, 943.6)	0.9106
5	49.46552	49	0.4545	1.010595	(49, 943.6)	0.4549
6	47.17980	49	0.5472	0.962749	(49, 943.6)	0.5476

Table A9: SVAR Baseline Coefficients for UK

Model: Ae = Bu where E[uu']=I

A =

1	0	0	0	0	0	0
C(1)	1	0	0	0	0	0
C(2)	C(5)	1	0	0	C(18)	0
0	C(6)	C(9)	1	C(15)	0	0
0	0	0	C(12)	1	C(19)	0
C(3)	C(7)	C(10)	C(13)	C(16)	1	0
C(4)	C(8)	C(11)	C(14)	C(17)	C(20)	1

B =

C(21)	0	0	0	0	0	0
0	C(22)	0	0	0	0	0
0	0	C(23)	0	0	0	0
0	0	0	C(24)	0	0	0
0	0	0	0	C(25)	0	0
0	0	0	0	0	C(26)	0
0	0	0	0	0	0	C(27)

	Coefficient	Std. Error	z-Statistic	Prob.
C(1)	-0.006776	0.004622	-1.465952	0.1427
C(2)	-0.036376	0.046742	-0.778229	0.4364
C(3)	0.086182	0.028638	3.009385	0.0026
C(4)	-0.039738	0.017202	-2.310124	0.0209
C(5)	-0.305129	0.414358	-0.736389	0.4615
C(6)	-0.050110	0.666550	-0.075179	0.9401
C(7)	0.356206	0.255261	1.395457	0.1629
C(8)	-0.079502	0.204373	-0.389004	0.6973
C(9)	54.49967	140.9851	0.386563	0.6991
C(10)	-2.692617	3.692019	-0.729307	0.4658
C(11)	1.421968	1.164029	1.221592	0.2219
C(12)	16.90056	6.488973	2.604504	0.0092
C(13)	5.147086	1.994213	2.581011	0.0099
C(14)	-0.320618	0.414588	-0.773342	0.4393
C(15)	-0.134169	0.254237	-0.527731	0.5977
C(16)	-0.298755	0.076539	-3.903311	0.0001
C(17)	0.014988	0.024435	0.613383	0.5396
C(18)	3.354332	4.185320	0.801452	0.4229
C(19)	-1.647727	0.864191	-1.906671	0.0566
C(20)	0.067147	0.234518	0.286319	0.7746
C(21)	0.192657	0.007800	24.69817	0.0000
C(22)	0.015552	0.000630	24.69818	0.0000
C(23)	0.046522	0.057958	0.802684	0.4222
C(24)	0.148033	0.379530	0.390042	0.6965
C(25)	0.180768	0.035345	5.114323	0.0000
C(26)	0.055313	0.014253	3.880843	0.0001
C(27)	0.054649	0.002213	24.69817	0.0000

Log likelihood	4854.480		
LR test for over-identification:			
Chi-square(1)	3.748148	Probability	0.0529

Table A10: SVAR baseline coefficients for Norway

Model: Ae = Bu where E[uu']=I

A =

1	0	0	0	0	0	0
C(1)	1	0	0	0	0	0
C(2)	C(5)	1	0	0	C(18)	0
0	C(6)	C(9)	1	C(15)	0	0
0	0	0	C(12)	1	C(19)	0
C(3)	C(7)	C(10)	C(13)	C(16)	1	0
C(4)	C(8)	C(11)	C(14)	C(17)	C(20)	1

B =

C(21)	0	0	0	0	0	0
0	C(22)	0	0	0	0	0
0	0	C(23)	0	0	0	0
0	0	0	C(24)	0	0	0
0	0	0	0	C(25)	0	0
0	0	0	0	0	C(26)	0
0	0	0	0	0	0	C(27)

	Coefficient	Std. Error	z-Statistic	Prob.
C(1)	0.005897	0.008033	0.734070	0.4629
C(2)	-0.003384	0.001495	-2.263376	0.0236
C(3)	-0.024341	0.011358	-2.143032	0.0321
C(4)	-0.015047	0.023920	-0.629054	0.5293
C(5)	-0.009820	0.010500	-0.935173	0.3497
C(6)	0.002114	0.021165	0.099887	0.9204
C(7)	-0.012732	0.058468	-0.217761	0.8276
C(8)	0.349865	0.166769	2.097907	0.0359
C(9)	-0.113671	0.138352	-0.821608	0.4113
C(10)	-0.372259	0.458647	-0.811645	0.4170
C(11)	4.069616	0.911550	4.464502	0.0000
C(12)	-5.268541	10.66028	-0.494221	0.6211
C(13)	-0.735906	1.002214	-0.734280	0.4628
C(14)	0.239763	0.475448	0.504289	0.6141
C(15)	0.022939	0.031639	0.725007	0.4684
C(16)	0.120498	0.074433	1.618882	0.1055
C(17)	0.029464	0.028040	1.050781	0.2934
C(18)	0.009369	0.020887	0.448545	0.6538
C(19)	-9.356509	6.821307	-1.371659	0.1702
C(20)	0.283563	0.228423	1.241393	0.2145
C(21)	0.192392	0.007790	24.69817	0.0000
C(22)	0.026991	0.001093	24.69817	0.0000
C(23)	0.004949	0.000201	24.59484	0.0000
C(24)	0.009964	0.001643	6.065494	0.0000
C(25)	0.266074	0.113204	2.350395	0.0188
C(26)	0.027288	0.008500	3.210327	0.0013
C(27)	0.078489	0.003178	24.69817	0.0000

Log likelihood	4146.465		
LR test for over-identification:			
Chi-square(1)	0.009529	Probability	0.9222

Table A11: Norway 12 lag roots

-0.868289	0.868289
0.638172 + 0.576243i	0.859837
0.638172 - 0.576243i	0.859837
-0.311290 + 0.798864i	0.857371
-0.311290 - 0.798864i	0.857371
-0.578830 - 0.628347i	0.854321
-0.578830 + 0.628347i	0.854321
0.850393 + 0.072151i	0.853448
0.850393 - 0.072151i	0.853448
-0.729583 + 0.435778i	0.849820
-0.729583 - 0.435778i	0.849820
0.190476 - 0.819713i	0.841552
0.190476 + 0.819713i	0.841552
0.297334 + 0.757178i	0.813466
0.297334 - 0.757178i	0.813466
0.654058 + 0.477527i	0.809830
0.654058 - 0.477527i	0.809830
-0.661067 + 0.445418i	0.797125
-0.661067 - 0.445418i	0.797125
-0.781552	0.781552
-0.731278 + 0.198765i	0.757810
-0.731278 - 0.198765i	0.757810
-0.008698 + 0.745888i	0.745939
-0.008698 - 0.745888i	0.745939
0.667930 - 0.072482i	0.671851
0.667930 + 0.072482i	0.671851
-0.627958 - 0.137981i	0.642938
-0.627958 + 0.137981i	0.642938
-0.255849 + 0.368105i	0.448285
-0.255849 - 0.368105i	0.448285
0.283391	0.283391

No root lies outside the unit circle.
VAR satisfies the stability condition.

Table A12: UK 12 Lag roots

0.858850 - 0.219157i	0.886371
-0.761503 + 0.424087i	0.871629
-0.761503 - 0.424087i	0.871629
0.116324 - 0.861859i	0.869674
0.116324 + 0.861859i	0.869674
0.826275 - 0.250359i	0.863372
0.826275 + 0.250359i	0.863372
0.849461 - 0.079448i	0.853168
0.849461 + 0.079448i	0.853168
0.521859 - 0.652607i	0.835603
0.521859 + 0.652607i	0.835603
0.416195 + 0.698133i	0.812778
0.416195 - 0.698133i	0.812778
-0.790742 - 0.187970i	0.812777
-0.790742 + 0.187970i	0.812777
0.748166 + 0.298813i	0.805631
0.748166 - 0.298813i	0.805631
-0.035741 - 0.804048i	0.804842
-0.035741 + 0.804048i	0.804842
0.693995 - 0.404368i	0.803208
0.693995 + 0.404368i	0.803208
-0.731694 - 0.310788i	0.794962
-0.731694 + 0.310788i	0.794962
-0.366103 + 0.600073i	0.702936
-0.366103 - 0.600073i	0.702936
0.504736 - 0.481263i	0.697404
0.504736 + 0.481263i	0.697404
0.684752	0.684752
-0.524015	0.524015
0.014595 + 0.449296i	0.449533
0.014595 - 0.449296i	0.449533

No root lies outside the unit circle.
VAR satisfies the stability condition.

Figure A3: Killian BSP VAR Norway

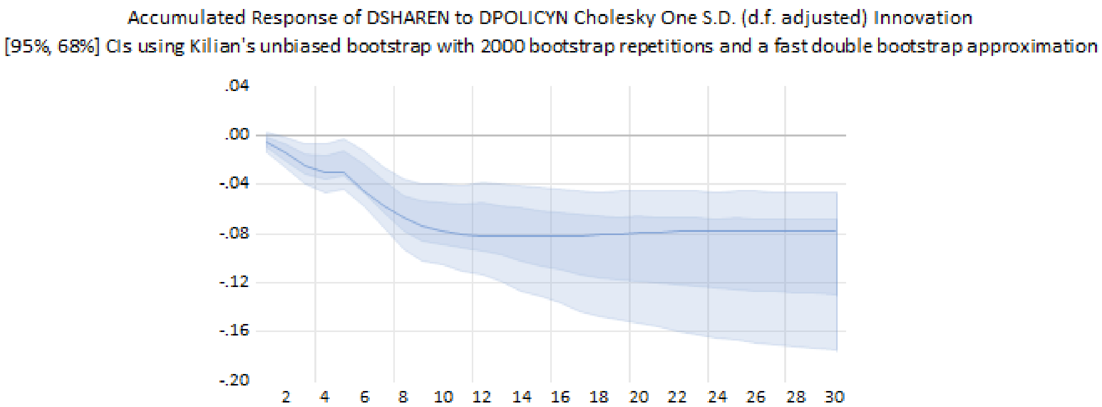

Figure A4: Killian BSP UK

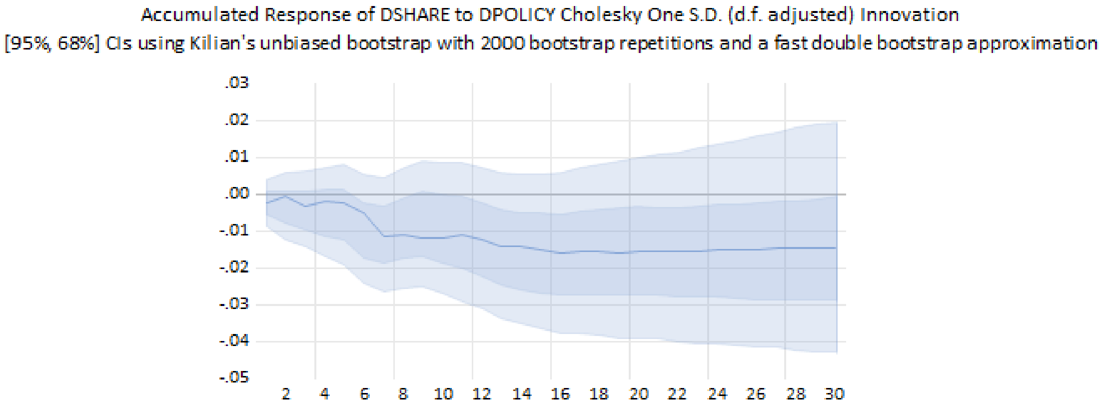

Figure A5: Norway Recursive Robustness with Brent OIL

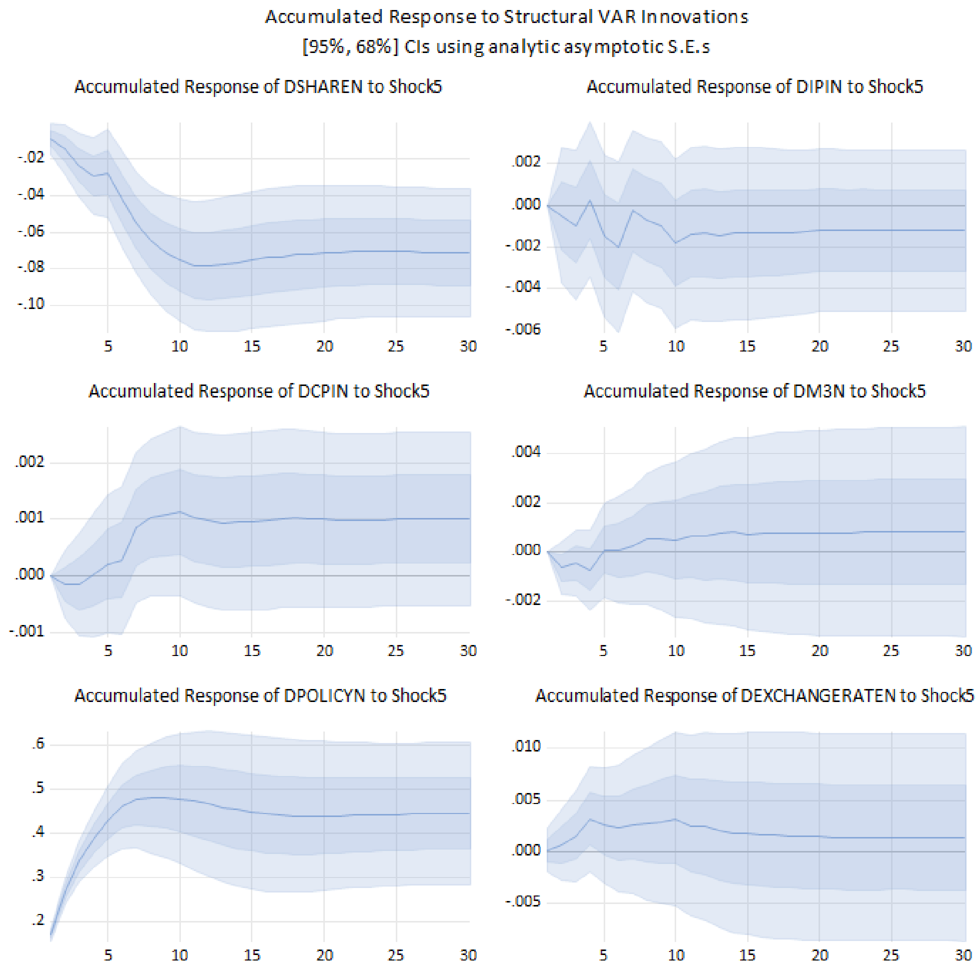

Table A13: SFEVD for Norway using Brent OIL

Period	S.E.	OIL	IPI	CPI	M3	POLICY	EXRATE	STOCKS
1	0.082589	4.878960	1.506662	6.252352	0.004247	1.175919	0.000346	86.18152
2	0.084021	4.914109	1.517333	6.878246	0.410336	1.646006	0.073349	84.56062
3	0.086151	4.848835	2.106377	9.465806	0.391475	2.619432	0.075756	80.49232
4	0.087377	4.729728	2.332052	9.203867	1.328365	2.942551	1.198065	78.26537
5	0.088328	4.857983	2.991397	9.095850	2.225677	2.914164	1.172591	76.74234
6	0.090074	4.690520	2.876607	8.749440	2.225221	5.254674	1.165288	75.03825
7	0.091790	4.519713	3.269004	9.257988	2.183777	7.028781	1.122154	72.61858
8	0.092432	4.460872	3.368151	9.130995	2.229928	8.031199	1.165428	71.61343
9	0.092754	4.437400	3.404313	9.110919	2.222978	8.527943	1.176748	71.11970
10	0.093103	4.519794	3.676764	9.109332	2.273553	8.652928	1.168007	70.59962
11	0.093206	4.549924	3.669034	9.167711	2.272386	8.727164	1.168930	70.44485
12	0.093243	4.548438	3.667409	9.166609	2.329805	8.720712	1.176906	70.39012
13	0.093303	4.579688	3.672222	9.154832	2.335198	8.727013	1.175740	70.35531
14	0.093354	4.578766	3.691128	9.146265	2.392318	8.721429	1.177094	70.29300
15	0.093394	4.576310	3.688977	9.180399	2.407779	8.736363	1.176203	70.23397
16	0.093437	4.572754	3.687921	9.210514	2.417300	8.754006	1.187312	70.17019
17	0.093458	4.572437	3.688994	9.235893	2.420642	8.753235	1.188899	70.13990
18	0.093467	4.572229	3.688262	9.237268	2.425151	8.758875	1.189337	70.12888
19	0.093472	4.571796	3.689820	9.237487	2.424948	8.760564	1.189938	70.12545
20	0.093476	4.571560	3.690905	9.240762	2.425338	8.762343	1.190014	70.11908
21	0.093479	4.571330	3.690783	9.242217	2.425411	8.764753	1.189957	70.11555
22	0.093480	4.571616	3.690679	9.242488	2.425814	8.765351	1.190332	70.11372
23	0.093481	4.571570	3.690694	9.242647	2.425827	8.765813	1.190371	70.11308
24	0.093481	4.571556	3.690707	9.242931	2.425842	8.765778	1.190391	70.11279

Figure A6: UK Recursive Robustness Brent OIL

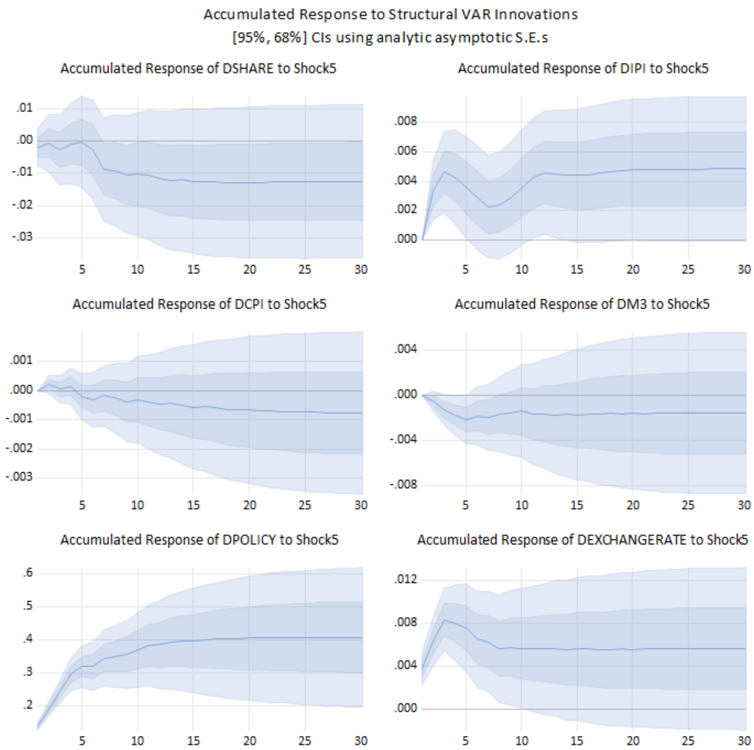

Table A14: UK SFEVD Brent Oil

Period	S.E.	OIL	IPI	CPI	M3	POLICY	EXRATE	STOCK
1	0.055353	4.579934	0.234638	0.867425	0.035240	0.111943	0.075129	94.09569
2	0.056068	4.482964	0.516639	1.059147	1.810149	0.161480	0.253474	91.71615
3	0.057222	5.857797	1.751550	1.017769	1.755621	0.294293	0.910240	88.41273
4	0.057502	5.912018	2.030196	1.260762	1.888251	0.400293	0.952403	87.55608
5	0.058446	7.934456	2.139025	1.335857	2.044328	0.397801	1.071991	85.07654
6	0.059395	7.834682	2.182505	3.495649	2.020075	0.551489	1.533092	82.38251
7	0.060016	7.679083	2.146383	3.561100	2.705016	1.567379	1.637562	80.70348
8	0.060271	8.087448	2.265521	3.568639	2.712208	1.556627	1.660894	80.14866
9	0.060356	8.064720	2.259125	3.663789	2.705748	1.606614	1.697653	80.00235
10	0.060426	8.167454	2.253921	3.699530	2.721869	1.604491	1.695635	79.85710
11	0.060589	8.196666	2.242309	3.747937	2.986406	1.601166	1.686526	79.53899
12	0.060716	8.162363	2.315447	3.856977	3.017807	1.621045	1.689871	79.33649
13	0.060786	8.189715	2.321273	3.860821	3.016465	1.629299	1.712638	79.26979
14	0.060814	8.202473	2.329852	3.884066	3.023954	1.629196	1.716488	79.21397
15	0.060832	8.201523	2.328542	3.884593	3.030220	1.634640	1.720378	79.20010
16	0.060847	8.220428	2.333305	3.883303	3.036462	1.634713	1.720515	79.17127
17	0.060865	8.219765	2.336202	3.897281	3.035524	1.633764	1.721932	79.15553
18	0.060873	8.225882	2.338346	3.897214	3.034913	1.633531	1.724020	79.14609
19	0.060877	8.231840	2.342136	3.896780	3.034650	1.633321	1.725872	79.13540
20	0.060879	8.231703	2.342914	3.898211	3.034504	1.633245	1.726365	79.13306
21	0.060881	8.233609	2.343032	3.897958	3.035244	1.633140	1.727286	79.12973
22	0.060882	8.234296	2.343245	3.897891	3.035330	1.633160	1.727503	79.12858
23	0.060883	8.234486	2.343524	3.897938	3.036217	1.633202	1.727638	79.12699
24	0.060883	8.235134	2.343674	3.897982	3.036886	1.633206	1.727644	79.12547

Figure A7: UK Baseline Robustness WIP

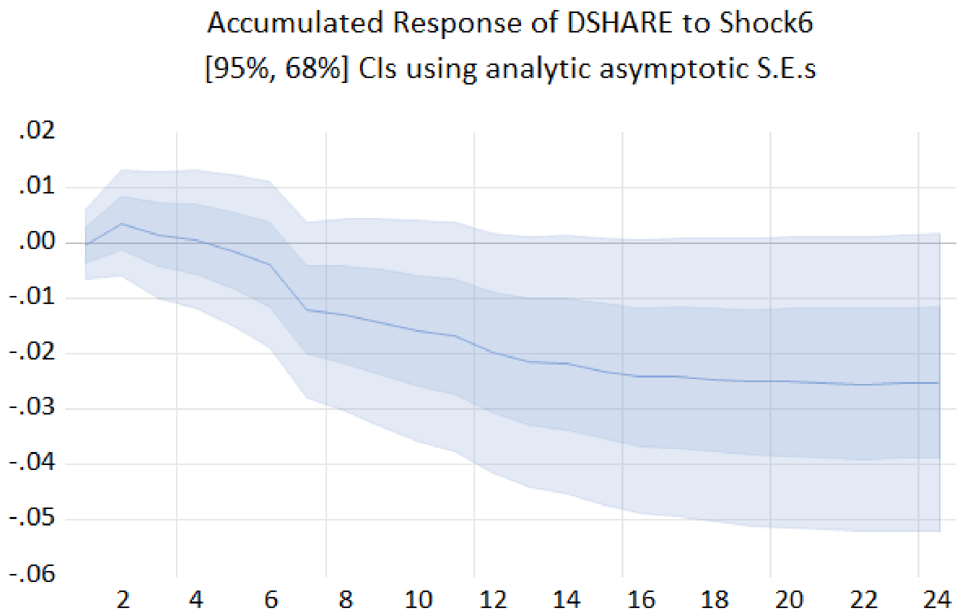

Table A15: UK SFEVD Baseline including WIP

Period	S.E.	Shock1	Shock2	Shock3	Shock4	Shock5	Shock6	Shock7	Shock 8
1	0.053949	0.114884	0.662140	0.069669	0.266481	0.418929	0.002637	0.541350	97.92391
2	0.054838	0.282920	1.692594	0.566788	0.387330	0.513763	0.489941	1.279328	94.78734
3	0.056767	2.502358	1.729924	2.455754	1.073859	0.564228	0.585794	1.305889	89.78219
4	0.057174	2.525521	2.359223	2.437575	1.060555	1.102855	0.604564	1.381530	88.52818
5	0.058318	2.750005	4.691016	2.777781	1.384860	1.080363	0.696825	1.462046	85.15710
6	0.059678	3.836305	4.534643	2.653972	2.094297	3.070816	0.843486	1.615810	81.35067
7	0.060663	3.859328	4.867691	2.746747	2.360122	3.123023	2.602772	1.614132	78.82618
8	0.060738	3.885855	4.901324	2.833124	2.401441	3.115641	2.618922	1.610974	78.63272
9	0.060993	3.873710	5.025389	2.809454	2.381468	3.376532	2.650631	1.722573	78.16024
10	0.061133	3.886187	5.002785	2.894375	2.407432	3.442479	2.703691	1.715266	77.94778
11	0.061237	3.922365	5.000288	2.885634	2.407596	3.518522	2.720892	1.759950	77.78475
12	0.061400	3.902645	4.989466	2.882042	2.395095	3.601279	2.916638	1.753160	77.55968
13	0.061538	3.925358	5.033189	2.890939	2.390067	3.641920	2.977701	1.794374	77.34645
14	0.061562	3.939441	5.029283	2.895465	2.393911	3.644161	2.981353	1.795815	77.32057
15	0.061590	3.943512	5.028613	2.895890	2.398815	3.648598	3.020204	1.794642	77.26973
16	0.061619	3.959599	5.025468	2.901878	2.397383	3.647193	3.048627	1.793582	77.22627
17	0.061642	3.967565	5.021861	2.910418	2.401928	3.655837	3.046386	1.795704	77.20030
18	0.061658	3.977050	5.019215	2.908940	2.404796	3.654137	3.051809	1.794776	77.18928
19	0.061666	3.988641	5.019919	2.913940	2.404520	3.653183	3.055499	1.794393	77.16990
20	0.061671	3.991006	5.019250	2.915516	2.406683	3.653064	3.055088	1.794754	77.16464
21	0.061675	3.993826	5.018943	2.915201	2.407032	3.656361	3.055736	1.794749	77.15815
22	0.061677	3.997209	5.018540	2.916151	2.406847	3.658711	3.055889	1.794649	77.15200
23	0.061678	3.997249	5.018384	2.916370	2.408102	3.658754	3.056728	1.794642	77.14977
24	0.061680	3.997452	5.018152	2.916334	2.408133	3.662700	3.056616	1.794553	77.14606

Figure A8: Norway Baseline Robustness WIP

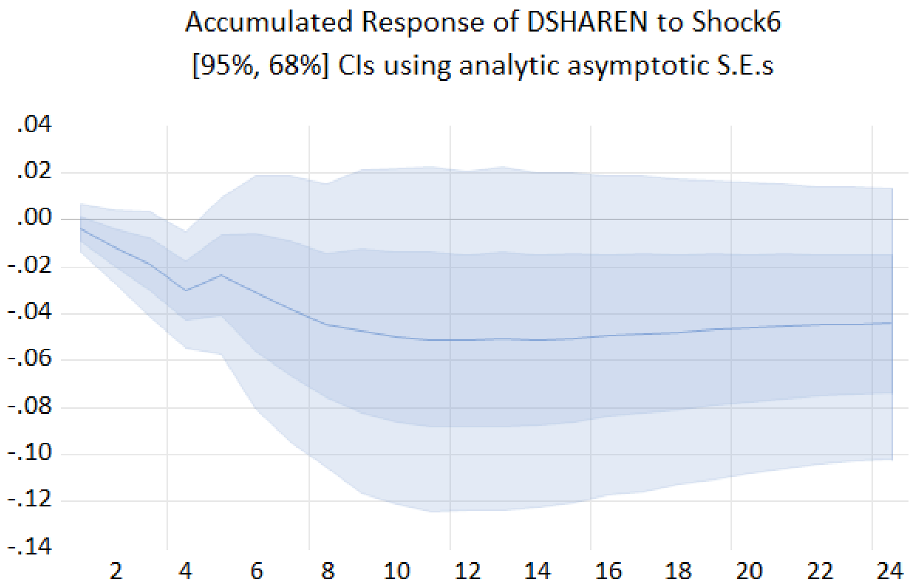

Table A16: Norway SFEVD Baseline Robustness WIP

Period	S.E.	Shock1	Shock2	Shock3	Shock4	Shock5	Shock6	Shock7	Shock 8
1	0.079546	0.005824	0.009964	2.038860	7.056468	0.118043	0.222087	0.159321	90.38943
2	0.081327	1.814182	0.168958	2.041065	7.296947	0.114810	1.127835	0.196109	87.24009
3	0.085801	5.247925	1.248178	2.457984	9.146710	1.402537	1.769278	0.179520	78.54787
4	0.086901	5.149230	1.480949	2.742181	8.928574	1.372276	3.342446	0.405659	76.57868
5	0.088052	5.020340	1.443402	3.592816	8.752103	1.833992	3.779264	0.486627	75.09146
6	0.090137	5.706499	1.380017	3.458992	8.360526	3.718383	4.238460	0.492210	72.64491
7	0.091511	5.573788	1.342102	4.150606	8.852544	3.967835	4.728742	0.561221	70.82316
8	0.092279	5.678128	1.577672	4.089977	8.717595	4.102785	5.223960	0.884039	69.72584
9	0.092772	5.619556	1.989740	4.047184	8.716626	4.381179	5.245099	0.875907	69.12471
10	0.093074	5.597857	2.097868	4.431366	8.688336	4.361527	5.269312	0.874425	68.67931
11	0.093429	5.589602	2.638820	4.398162	8.737683	4.332029	5.251137	0.892361	68.16021
12	0.093557	5.603404	2.780538	4.387910	8.715619	4.372754	5.238295	0.913220	67.98826
13	0.093737	5.581918	2.947360	4.379379	8.682166	4.366128	5.224947	1.000831	67.81727
14	0.093834	5.573362	3.039013	4.375251	8.681417	4.403132	5.215632	0.998789	67.71340
15	0.093943	5.571813	3.067934	4.369569	8.745841	4.441808	5.210997	1.014756	67.57728
16	0.094049	5.559374	3.138223	4.360653	8.786440	4.494116	5.210459	1.012988	67.43775
17	0.094098	5.553546	3.150453	4.356240	8.828619	4.515479	5.212547	1.014591	67.36852
18	0.094146	5.550191	3.167644	4.352241	8.827870	4.575017	5.212867	1.013872	67.30030
19	0.094174	5.548824	3.180938	4.350722	8.822812	4.603530	5.218503	1.015198	67.25947
20	0.094202	5.545795	3.180889	4.348220	8.818363	4.648459	5.220971	1.016858	67.22045
21	0.094224	5.543246	3.181692	4.346179	8.815248	4.678955	5.225089	1.017888	67.19170
22	0.094242	5.541453	3.181063	4.345255	8.812548	4.708641	5.225940	1.017521	67.16758
23	0.094254	5.540018	3.180439	4.344868	8.811593	4.727110	5.227071	1.017586	67.15132
24	0.094262	5.539427	3.181483	4.344638	8.811262	4.737119	5.227018	1.017630	67.14142

END.